
CEI 0-16 Central Plant Controller — Master Reference

HiTEKS CCI Programme · Phase 1 documentation baseline

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Lead path: Toradex Verdin Development Board **99991105** + SoM **0063** (i.MX8M Plus Quad 4GB IT, no Wi-Fi). Eval kit is BSP/lab only — field CCI requires custom carrier for C1 remainder, C2, C4, C5, C10.

Living document: edit canonical Markdown under `knowledge-base/` and programme root, then run `scripts/build-master-reference.ps1` to regenerate this PDF. Do not edit the generated `.tex` by hand.

Target product	CEI 0-16 Annex O/T/M Central Plant Controller (100 kW–6 MW MT)
Phase 1 budget	~€1,000 prototype (single unit)
Lead vendor score	Toradex 84 · Variscite 67 · NXP 60 (post Wave 1)
RAG corpus	Telemetry <code>ingest-ccli.ps1</code> · hybrid vector + keyword + graph
Companion site	<code>website/</code> (Chapters 01–10)

Included in this edition

Programme roadmap · KB structure · vendor C1–C10 matrix · Wave 1 scorecard · kit catalog · prototype BOM & missing-doc index · regulations classification · RAG architecture · download index.

Contents

I	Programme overview	6
1	Project roadmap	7
1.1	Original CCI-Class Controller - Project Roadmap	7
1.1.1	System architecture (original design)	7
1.1.2	Compute module: SoM + custom carrier	8
1.1.3	Prototype BOM target (single unit, ~€1,000)	9
1.1.4	Path to Gerbers	10
1.1.5	Firmware architecture & language choice	10
1.1.6	Timeline - Phase 1: Hardware + firmware prototype (~10-12 weeks, balanced pace)	11
1.1.7	Phase 2: Certification path (separate track, 9-18 months)	11
1.1.8	Team/skills needed	11
1.1.9	To do	12
2	Knowledge base brief	13
2.0.1	Knowledge Base tree	13
2.0.2	Branch briefs	13
2.0.3	Mapping summary (quick view)	17
2.0.4	Recommended folder layout (on disk -> RAG)	17
2.0.5	To do	18
2.0.6	Telemetry RAG source_ids already useful for this KB	19
2.0.7	Keywords (routing)	19
II	Vendor selection & prototype BOM	20
3	Vendor selection (C1-C10)	21
3.1	CCI Vendor Selection - Product Analysis & Dev-Kit Matrix	21
3.1.1	Decision rule (read first)	21
3.1.2	Table A - CCI control checklist (master requirements)	22
3.1.3	Table B - Vendor product shortlist (analyze each company)	23
3.1.4	Toradex kit row - Verdin Development Board (PN 99991105)	23
3.1.5	Table C - SoM silicon shortlist (summary; detail in SoM comparison)	26
3.1.6	Table D - Phase 1 kit recommendation	26
3.1.7	Scorecard template (fill when datasheets arrive)	26

3.1.8	Toradex download paths (local + URL)	27
3.1.9	To do	27
3.1.10	Keywords (RAG)	27
4	Vendor decision scorecard	28
4.1	CCI Vendor Decision Scorecard (Phase 1) - post Wave 1 ingest	28
4.1.1	Wave 1 ingest (2026-07-25)	28
4.1.2	Evidence gates	28
4.1.3	Ranking (after Wave 1)	29
5	Vendor kit catalog	30
5.1	CCI Vendor Kit Catalog (Prototype BOM)	30
5.1.1	Lead pick (Phase 1)	30
5.1.2	Supplier categories	30
5.1.3	Custom carrier (after SoM lock)	31
6	Prototype BOM & missing index	32
6.1	CCI Prototype BOM & Missing Document Index	32
6.1.1	Snapshot - downloaded vs still missing	32
6.1.2	Document gate (must exist before schematic freeze)	34
6.1.3	Architecture & system (non-PCB but BOM-driving)	36
6.1.4	Compute - SoM + eval (Phase 1 buy)	36
6.1.5	Ethernet & plant networking (C1)	37
6.1.6	Serial / Modbus plant (C2) - signals	37
6.1.7	Digital I/O - DI signals & DO relays (C4)	37
6.1.8	Time sync - GNSS (C5)	37
6.1.9	Cellular - LTE (optional rev A)	38
6.1.10	Security / trust (C6)	38
6.1.11	Power (C10)	38
6.1.12	Mechanical / HMI / misc	38
6.1.13	Software / stack (not PCB BOM, but prototype “bill”)	39
6.1.14	To do	39
6.1.15	RAG / folder convention	39
6.1.16	Keywords	39
III	Regulations & protocols	40
7	Module regulations classification	41
7.1	CCI Module - Hardware/Protocol Inventory & Regulations Classification	41
7.2	Modulo CCI - Inventario HW/protocolli e classificazione normative	41
7.2.1	Italian <-> English glossary / Glossario IT <-> EN	41
7.2.2	Module inventory (source description)	42
7.2.3	Inventario modulo (descrizione sorgente)	42

7.2.4	Regulation taxonomy (RAG classes)	44
7.2.5	Tassonomia normative (classi RAG)	45
7.2.6	Classification table - every cited norm	45
7.2.7	Tabella di classificazione - ogni norma citata	46
7.2.8	Feature -> regulation map (for RAG queries)	47
7.2.9	Mappa funzione -> normativa (per query RAG)	47
7.2.10	RAG routing packs (copy into chat / query filters)	47
7.2.11	Pacchetti di routing RAG (da usare in chat / filtri query)	47
7.2.12	To do (regulation PDFs) / Backlog acquisizione (P0 first)	48
7.2.13	Notes / caveats / Note	48
IV	Engineering infrastructure	49
8	Industrial RAG architecture	50
8.1	CCI Industrial RAG Architecture (three indexes)	50
8.1.1	Three indexes	50
8.1.2	Artifacts	51
8.1.3	Edge types (seed graph)	51
8.1.4	Query contract (already on /v1/query)	51
8.1.5	Apply after code deploy	51
8.1.6	Still out of MVP (next)	52
8.1.7	Keywords	52
9	Master download index	53
9.1	CCI Knowledge Base - Master Download Index	53
9.1.1	Yocto / BitBake (01-yocto/, 02-bitbake/)	53
9.1.2	GitHub references (01-yocto/github-refs/)	54
9.1.3	Engineering / Toradex kit (08-engineering/)	54
9.1.4	3b. NXP SoC datasheets (08-engineering/reference/)	55
9.1.5	Protocols / EU regulations (07-protocols/)	55
9.1.6	Repo-root project docs (mirrored to Telemetry projects/ccli/)	55
9.1.7	To do (downloads)	56
9.1.8	Re-ingest helper (Telemetry)	56
V	Validation & test	57
10	PF2 validation & mocking strategy	58
10.1	CCI Validation & Mocking Strategy - PF2 Bench Ladder	58
10.1.1	Recommended progression	58
10.1.2	Level 1 - software-only mocking	58
10.1.3	Level 2 - physical-interface HIL (recommended)	60
10.1.4	Level 3 - real energy analyzer	60

10.1.5	Level 4 - real-time power-system HIL	61
10.1.6	Digital-input test box (C4)	61
10.1.7	Dry-contact output tests (C4)	61
10.1.8	IEC 61850 and IEC 104 bench	62
10.1.9	Cross-interface failure matrix	62
10.1.10	Practical Phase 1 PF2 bench	62
10.1.11	Test evidence required	63
10.1.12	To do	63
10.1.13	RAG keywords	64
11	Mocking / PF2 bench BOM	65
11.1	CCI Mocking / PF2 Bench BOM	65
11.1.1	Buy waves (recommended)	65
11.1.2	Topology (what this BOM builds)	66
11.1.3	Wave A - L1 software mock + L2 interface HIL (BUY NOW)	66
11.1.4	Wave B - DI test box + DO lamps (BUY NEXT)	67
11.1.5	Wave C - L3 real energy analyzer (BUY NEXT before field pilot)	68
11.1.6	Wave D - L4 power-system HIL (DEFER)	69
11.1.7	Controller under test (already on product BOM)	69
11.1.8	Consolidated buy list (print / purchase)	69
11.1.9	Cost roll-up	70
11.1.10	Documents still required for this bench	71
11.1.11	Mapping to C1-C10 / validation levels	71
11.1.12	To do	71
11.1.13	Keywords	71
VI	RAG knowledge-base	72
12	RAG knowledge-base (clustered inventory)	73
12.1	CCI RAG Knowledge Base - Clustered Inventory	73
12.1.1	Status rollup (document gate D0-D21)	73
12.1.2	Cluster A - Programme & architecture (ready)	74
12.1.3	Cluster B - Architecture freeze blockers (P0)	74
12.1.4	Cluster C - SoM & silicon	74
12.1.5	Cluster D - Carrier electronics (MPN TBD)	75
12.1.6	Cluster E - Plant & protocols	75
12.1.7	Cluster F - Optional / rev A	75
12.1.8	Cluster G - Fab & mechanical	75
12.1.9	KB branch clusters (RAG tree)	76
12.1.10	GitHub / layer-pin clusters (SHA policy)	76
12.1.11	Ingest clusters (Telemetry)	77
12.1.12	To do (by cluster priority)	77
12.1.13	Keywords	77

13 GitHub paths & layer pin catalog	78
13.1 CCI GitHub Paths & Layer Pin Catalog	78
13.1.1 SHA / hash recording template (Architecture Freeze)	78
13.1.2 Cluster A - Already harvested (github-refs/ + RAG)	79
13.1.3 Cluster B - Toradex lead path (PN 99991105 + SoM 0063)	79
13.1.4 Cluster C - NXP silicon reference (all vendors)	80
13.1.5 Cluster D - Variscite alternate (DART-MX8M-PLUS)	80
13.1.6 Cluster E - Compulab alternate / study	80
13.1.7 Cluster F - Yocto core & OE (reference - usually pinned via manifest)	80
13.1.8 Cluster G - Security, TPM, IMA, and image integrity	81
13.1.9 Cluster H - Application / protocol (BitBake sources)	81
13.1.10 Cluster I - Bench / lab tools (non-Yocto)	81
13.1.11 Quick lookup - lead path minimum pin set	81
13.1.12 To do	82
13.1.13 Keywords	82

I Programme overview

Chapter 1

Project roadmap

1.1 Original CCI-Class Controller - Project Roadmap

- **Target:** CEI 0-16 Annex O/T/M compliant Central Plant Controller, flexible 100 kW-6 MW MT plant range
- **Scope:** Original hardware/firmware design (not a clone of any existing commercial product)
- **Prototype budget:** ~€1,000 (single unit, BOM + fab + assembly)
- **Timeline stance:** Balanced - solid working prototype in ~10-12 weeks; grid/cyber certification is a separate, longer track
- **Existing certification:** organization already holds **ATEX** (EN/IEC 60079) - reuse QMS, Notified Body relationship, and technical-file practice for the CEI 0-16 track.
- **Alternate track:** see companion brief **CCLI-CLONE-RE-001** (CCI_Clone_RE_Option.pdf) for the clone / reverse-engineering option, risks, and decision guide.

OK **Already in place:** your company is **ATEX-certified**. Standard MT CCI cabinets are usually *non-hazardous area* - confirm per project whether Ex marking is required on the CCI itself.

! **Set expectations up front:** the hardware prototype is the *cheap and fast* part. Remaining certification for this product (IEC 62443, IEC 61850, IEC 62351, CEI 0-16 demo) is realistically **€45,000-110,000+** and 9-18 months. ATEX is not budgeted again here.

1.1.1 System architecture (original design)

This block diagram reflects the *generic, publicly-documented functional interfaces* that CEI 0-16 Annexes O/T require of any CCI (dual DSO/Operator Ethernet separation, plant-side ports, serial link to the energy analyzer, time sync, DI/DO for status/command) - not any specific vendor's schematic or component selection. See **CCI_Architecture.png** (primary visual) and **CCI_Architecture.mermaid** (editable companion). **Core functional blocks:**

Block	Function	Notes / track
Design approach	Option A (default): original SoM + carrier from public CEI 0-16 interfaces	Primary track - independent schematic, layout, firmware
Design approach	Option B: clone via reverse engineering (AiLux-class reference)	See CCLI-CLONE-RE-001 (CCI_Clone_RE_Option.pdf) - higher IP/legal risk; gated research spike only
Compute module (SoM)	Runs embedded Linux, IEC 61850 server, Modbus master, PF2 control logic	Option A: Toradex/Variscite/Compulab-class. See Sec.2
Ethernet interfaces (>=4 port managed switch or SoM+switch IC)	Eth_A -> DSO; Eth_B -> Operator; 2x plant ports	DSO/plant isolation required - two isolated PHYs/VLANs minimum
RS-485/RS-232 serial (x2, isolated)	Modbus RTU master to energy analyzer; IEC 60870-5-101 optional	Opto-isolated; map registers from datasheet
GNSS module	Time sync for timestamped measurements/events	u-blox-class module + antenna
Cellular modem (LTE Cat-1/Cat-4)	Remote config/monitoring, backup comms	Optional on rev A if DSO link is wired
Digital I/O	DI: 10-120 Vdc status; DO: dry contact curtailment relay	Critical PF2 path - re-implement on Option B, do not copy reference network
Secure element + secure boot	Root of trust, key storage, signed firmware	Option A: your key hierarchy; Option B: never reuse reference keys/boot chain
Local console	USB service port for commissioning/diagnostics	Outside certified control path
Power input	12-24 Vdc wide-range isolated DC-DC	Box CCI UPS/battery context

1.1.2 Compute module: SoM + custom carrier

Decision (from our discussion): off-the-shelf industrial SoM, original carrier board design. Candidate SoM families to evaluate (pick based on Linux BSP maturity, industrial temp range, and 10-year longevity commitment - you need this for a utility-facing product):

- Toradex Verdin iMX8M Mini/Plus
- Variscite DART-MX8M-Mini / DART-MX8M-Plus
- Compulab CL-SOM-iMX8
- NXP i.MX8M Mini EVK-class reference designs

All of these ship a maintained Yocto BSP, handle DDR/power-sequencing/thermal internally, and let your carrier board stay a straightforward 2-4 layer design (Ethernet PHYs, RS-485 transceivers, GNSS/LTE module sockets, DI/DO conditioning, power).

1.1.3 Prototype BOM target (single unit, ~€1,000)

Item	Est. cost (EUR)	Notes
Industrial SoM (qty 1)	€60-120	Toradex/Variscite-class, i.MX8M Mini tier
Carrier PCB fab (4-layer, small batch 5-10 pcs)	€50-150	JLCPCB/PCBWay prototype tier
Assembly (self or light-touch SMT service)	€100-250	BGA stays on the SoM module - carrier assembly is hand/reflow-feasible
Ethernet PHYs/magnetics/RJ45 (x4)	€15-25	
RS-485/RS-232 isolated transceivers (x2)	€10-15	
GNSS module + antenna	€25-40	u-blox-class
LTE module + antenna + SIM holder	€25-40	Quectel-class; optional on rev A
Secure element	€2-5	e.g. Microchip ATECC608B / STSAFE-A110 class part
DI/DO conditioning (opto-isolators, relays)	€15-25	
Power stage (wide-range isolated DC-DC)	€15-25	
DIN rail hardware, connectors, terminal blocks, enclosure	€40-60	
Passives, stencil, shipping, contingency	€80-150	
Total (single prototype)	~€440-900	Leaves headroom within €1,000 for one respin's worth of parts

This is a *component-cost* estimate for one unit at prototype quantities - not a manufactured unit cost at volume, which would drop significantly (likely €150-300/unit at 100+ qty once NRE is amortized).

1.1.4 Path to Gerbers

#	Step	Activity (Altium Designer)
1	Architecture freeze	Lock interfaces, connector count, power budget before schematic work
2	Schematic capture	Altium Designer - build from datasheets; link SoM/PHY foot-prints early
3	Design review	Power sequencing, ESD/surge on field I/O, isolation barriers
4	PCB layout	SoM first, then Ethernet PHY/magnetics, RS-485, power; keep analog away from switching/RF
5	DFM check	Altium DRC + fab DFM portal (JLCPCB/PCBWay) against stackup rules
6	Fab outputs	Altium Output Job : Gerber, drill, pick-and-place, BOM (Active-BOM)
7	Fab + assembly	1-3 week prototype turnaround
8	Bring-up	Power-on, boot SoM, verify each interface before firmware integration

1.1.5 Firmware architecture & language choice

- **OS/BSP:** Yocto-based embedded Linux (matches SoM vendor's supported BSP) - gives you package management, OTA update tooling, and a known-good foundation auditors are familiar with.
- **Core control/protocol path (IEC 61850 server, Modbus master, PF2 curtailment logic, GOOSE handling):** C/C++. This is the code that will eventually face conformance and security audits - auditors and test labs are far more comfortable reviewing statically-typed, analyzable C/C++ than dynamic scripting languages, and real-time GOOSE timing requirements favor it anyway.
- **Config/commissioning tooling, local web console, diagnostics:** Python or a lightweight web stack is fine here - this code is outside the certified control-path boundary, so language choice is a productivity call, not a compliance one. Keep the boundary between "certified path" and "tooling" architecturally clean from day one - it makes scoping the IEC 62443-4-2 assessment much cheaper later.
- **IEC 61850 stack:** start on [libiec61850](https://github.com/mz-automation/libiec61850) (MZ Automation) for prototype/demo - **GPLv3** open-source core (MMS, GOOSE, SV, R-Session); **commercial licenses sold separately** by MZ Automation for shipping a closed product. Treat OSS as *free to prototype, pay to ship* - budget the commercial license from day one. Evaluate commercial stacks (SISCO, Triangle MicroWorks) before paid conformance testing - see Sec.7.
- **Security baseline to build in from day one** (cheap now, expensive to retrofit later): secure boot chain, signed firmware images, TLS/DTLS for IEC 62351, key storage in the secure element, basic RBAC on the local console, audit logging. IEC 62443-4-1 specifically audits *how* you built and maintain the product (your SDL), not just what features it has - so start documenting your development process now, not after the fact.

1.1.6 Timeline - Phase 1: Hardware + firmware prototype (~10-12 weeks, balanced pace)

Weeks	Milestone
1-2	Architecture freeze, SoM selection finalized, schematic capture starts
2-4	Schematic complete + design review, long-lead parts ordered (SoM, LTE/GNSS modules)
4-6	PCB layout, DFM check, Gerbers to fab
6-8	Fab + assembly turnaround
8-10	Hardware bring-up: power, boot, verify each interface (Ethernet x4, RS-485 x2, GNSS, LTE, DI/DO)
10-12	Firmware integration: libiec61850 server running, Modbus master polling a real analyzer, DO-driven curtailment stub, local console reachable

Exit criteria for this phase: a DIN-rail prototype that boots reliably, exposes an IEC 61850 MMS server over Eth_A, polls a Modbus energy analyzer over RS-485, timestamps data via GNSS, and can toggle a DO relay from a simulated DSO command. This is a credible demo/investor/pilot-partner artifact - it is *not* yet field-deployable at a real MT POC.

1.1.7 Phase 2: Certification path (separate track, 9-18 months)

Scope note: ATEX (EN/IEC 60079) is **already held** - not a new line item. Table covers remaining CEI/grid/cyber work.

Workstream	What it involves	Rough cost	Duration
ATEX (EN/IEC 60079)	Already certified - reuse NB/QMS/technical-file process	Sunk / reuse	-
IEC 62443-4-1	Secure SDLC process audit (partial overlap with ATEX QMS)	€10,000-30,000	2-5 months
IEC 62443-4-2	Device security lab test (SL1 min, target SL2)	€15,000-40,000	2-4 months
IEC 61850 conformance	MMS/GOOSE/reporting lab test	€10,000-25,000	1-3 months
IEC 62351	Cybersecurity protocol testing	Overlaps 4-2	3-6 months
CEI 0-16 Annex O/T + DSO demo	PF2/observability validation	Integrator time	1-3 months

All-in for remaining CEI-class CCI certification: roughly €45,000-110,000 over 9-18 months (ATEX excluded).

1.1.8 Team/skills needed

- Embedded hardware engineer (schematic + layout, comfortable with SoM carrier design)
- Embedded Linux/firmware engineer (Yocto, C/C++, protocol stacks)
- Someone who can own the IEC 62443-4-1 process documentation - leverage existing ATEX technical-file and NB workflow

- Accredited test lab relationship early - your ATEX Notified Body may partner on 62443/61850 scoping
-

1.1.9 To do

1. Confirm SoM family choice (Toradex vs Variscite vs Compulab) based on your BSP/tooling comfort and long-term sourcing preference.
2. I can draft the interface list and pin budget (connector-by-connector) as a starting schematic outline - original to your design, not copied from any existing product.
3. Get a scoping quote from one IEC 62443/61850 test lab now, even before hardware exists - their capacity and requirements should shape your firmware architecture decisions, not the other way around.

Happy to start on the schematic-level interface/pin-budget document next, or go deeper on any single phase above.

Chapter 2

Knowledge base brief

CCI Knowledge Base Brief - Identify, Structure, RAG **Document ID:** CCLI-KB-001 **Revision:** 1.3 **Date:** 2026-07-25 **RAG source_id:** ccli-knowledge-base-brief **Purpose:** Define the Knowledge Base tree for the Original CCI project, map each branch to existing Telemetry RAG sources, and list what still must be collected and ingested. **On-disk root:** c:\Yahia\projects\CCLI\knowledge-base\ - mirrored in Telemetry under documents/projects/ccli/ for RAG ingest. **Master download paths:** DOWNLOADS.md (local path <-> URL <-> source_id).

2.0.1 Knowledge Base tree

Knowledge Base
 "œ"œ"œ Yocto Docs
 "œ"œ"œ BitBake Docs
 "œ"œ"œ Custom Layers
 "œ"œ"œ Recipes (.bb)
 "œ"œ"œ Kernel & Device Trees
 "œ"œ"œ Application Source Code
 "œ"œ"œ Protocol Documentation
 "a"œ"œ Engineering Knowledge Base

Each branch below: **what it is, CCI relevance, already in RAG, missing / to add.**

2.0.2 Branch briefs

2.1 Yocto Docs

What: Upstream and vendor Yocto Project documentation - concepts, image build flow, SDK, licenses, reproducible builds. **CCI use:** Build Embedded Linux for the chosen SoM (Toradex / Variscite / Compulab / NXP); OTA/package baseline; audit trail for IEC 62443-4-1.

Item	Status
Upstream Yocto Mega-Manual / Overview (version pinned to SoM BSP)	INGESTED - ccli-yocto-overview <- 01-yocto/yocto-5.0.13-theyoctoproject.pdf (Scarthgap 5.0.13)
Vendor Yocto getting-started (chosen SoM)	UNBLOCKED - Toradex kit docs ON DISK; SoM SKU still to lock - 08-engineering/CCI_Vendor_Selection.md
NXP i.MX Yocto Project User's Guide	PARTIAL - ccli-nxp-yocto-ug <- text extract; drop official PDF at 01-yocto/NXP_UG10164_iMX_Yocto_Project_Users_Guide.pdf
GitHub Yocto / stack refs	INGESTED - ccli-github-yocto-refs
Local notes (ccli-yocto-bsp-notes)	INGESTED
Vendor selection / eval kits	INGESTED - ccli-vendor-selection + Toradex PDFs

Download paths: DOWNLOADS.md · 01-yocto/DOWNLOADS.md

2.2 BitBake Docs

What: BitBake user manual - tasks, variables, do_fetch/do_compile/do_install, overrides, sstate, dependency graph. **CCI use:** Maintain meta-ccli recipes without breaking vendor layers; debug failed builds; document build process for certification.

Item	Status
BitBake User Manual (same Yocto release as BSP)	INGESTED - ccli-bitbake-manual <- 02-bitbake/bitbake-user-manual-scarthgap-era-extract.md
Internal "how we build CCI images" runbook	MISSING (create after first successful image)

Suggested RAG ids: ccli-bitbake-manual (*this download*), ccli-build-runbook

2.3 Custom Layers

What: Project-owned OpenEmbedded layers (e.g. meta-ccli) sitting on top of vendor + Poky/OE-core. **CCI use:** Carrier board machine config, Distro features (secure boot, firewall), packagegroups for certified vs tooling paths.

Layer / artifact	Status
meta-ccli layer README + architecture note	MISSING - create at Architecture Freeze
Distro config (ccli.conf) - Eth_A/B policy, signed images	MISSING
Vendor layer pin list (git SHAs / release tags)	MISSING

Suggested RAG ids: ccli-meta-ccli-readme, ccli-layer-pinlist **Brief content to capture later:**

- Layer priority / BBLAYERS
- What is vendor vs what is ours
- Certified-path packagegroup vs diagnostics packagegroup

2.4 Recipes (.bb / .bbappend)

What: BitBake recipes for apps, libraries, kernel modules, configs, and .bbappend overrides of vendor packages. **CCI use:** IEC 61850 stack packaging, Modbus master, PF2 logic, secure-element tools, local console packages.

Recipe family	Status
libiec61850 (or commercial stack) recipe	MISSING
CCI control app recipe	MISSING
Secure-element / HAB helpers	MISSING
.bbappend for firewall, users, ssh hardening	MISSING

Suggested RAG ids: ccli-recipes-index, per-recipe ccli-bb-<name> **Brief rule:** every recipe ingested as markdown should include: SUMMARY, LICENSE, depends, install paths, and whether it is inside or outside the certified control path.

2.5 Kernel & Device Trees

What: Linux kernel version from BSP, out-of-tree modules if any, device-tree (.dts / .dtsi) and carrier overlays. **CCI use:** Ethernet switch / PHYs, RS-485 UARTs, DI/DO GPIO, GNSS UART, LTE USB/PCIe, secure element I2C/SPI, GNSS PPS if used.

Artifact	Status
Vendor kernel + DT for chosen SoM	MISSING - after SoM pick
Carrier overlay imx8m*-ccli-carrier.dts	MISSING
Pin mux / connector map (from Altium)	MISSING - next engineering deliverable
NXP SoC datasheet (electrical)	INGESTED - ccli-nxp-imx8m-dqlq-cec (+ precise extract)

Suggested RAG ids: ccli-dt-carrier, ccli-kernel-config, ccli-pinmux-map

2.6 Application Source Code

What: CCI user-space and libraries - IEC 61850 server, Modbus master, PF2 curtailment, config/commissioning tools, diagnostics. **CCI use:** Core product behavior; conformance and security audit surface.

Component	Language (roadmap)	Status
IEC 61850 / GOOSE / MMS path	C/C++	NOT STARTED
Modbus RTU master	C/C++	NOT STARTED
PF2 curtailment + DI/DO	C/C++	NOT STARTED
Local web / commissioning	Python/web OK	NOT STARTED (outside certified path)
Architecture / firmware notes	docs	INGESTED - roadmap, SoM comparison, clone RE brief

Suggested RAG ids: ccli-app-iec61850, ccli-app-modbus, ccli-app-pf2, ccli-app-tooling **Brief rule:** ingest design docs + API/ICD notes early; ingest source trees as structured markdown indexes (file map + responsibilities), not raw multi-MB dumps.

2.7 Protocol Documentation

What: Grid and plant protocols, cybersecurity standards for protocols, and regulatory classification for the AiLux-class CCI module.

Protocol / standard	Role	RAG status
CEI 0-16 Annex O	CCI / DER plant controller	INGESTED cei-0-16-allegato-o
CEI 0-16 Annex T	IEC 61850 information exchange	INGESTED cei-0-16-allegato-t
AiLux CCI manual (reference only)	Behavior reference - do not clone	INGESTED cei-0-16-cci-manual-ailux
Module regs classification (IT+EN)	Norm -> class -> source_id routing map	INGESTED ccli-cci-module-regs-class - 07-protocols/CCI_Module_Regulations_Clas
Directive (EU) 2018/2001 RED II	Energy policy context	INGESTED eu-2018-2001-red
Regulation (EU) 2024/2847 CRA	Product cyber (clone path)	INGESTED eu-2024-2847-cra
ARERA 540/2021	Italian market / CCI mandate	MISSING -> target arera-540-2021
CEI EN 61557-12:2018	Metrology / monitoring	MISSING -> target cei-en-61557-12-2018
IEC 62443-4-1 / 4-2	IACS cyber process & component	MISSING
IEC 62351-3 6 (and -7/-8/-9/-14)	Secure protocols	MISSING - see classification doc Sec.3
IEC 61850 core (MMS/GOOSE)	DSO Eth_A	PARTIAL via Annex T; full standard MISSING
Modbus RTU/TCP	Energy analyzer / plant	Analyzer map MISSING ; gateway refs INGESTED (ccli-igate-850-manual, ccli-moxa-mgate-5119)
IEC 60870-5-101/103/104	Serial / Ethernet telecontrol	MISSING
FIPS 140-2	TPM / crypto module claim	MISSING
IEEE 1588 / GNSS time	Timestamping	SoC feature in NXP DS; GNSS module DS MISSING

RAG routing packs (from classification doc): Pack A (Italian CCI), B (cyber partial), C (protocols partial), D (market/metrology - blocked). **Suggested RAG ids when adding:** ccli-iec61850-overview, ccli-modbus-analyzer-map, iec-62351-*, arera-540-2021

2.8 Engineering Knowledge Base

What: Cross-cutting project knowledge - architecture, BOM, certification, SoM selection, Altium flow, test, decisions.

Topic	RAG status
Project roadmap (Option A)	INGESTED ccli-project-roadmap
Architecture diagram (PNG)	File in repo; describe in roadmap
Clone / RE option (Option B)	INGESTED ccli-clone-re-option
SoM / SoC comparison	INGESTED ccli-som-comparison
Vendor selection + Toradex kit	INGESTED ccli-vendor-selection + Toradex datasheet/design-guide/family PDFs in 08-engineering/
Master download index	knowledge-base/DOWNLOADS.md
NXP i.MX 8M Dual/Quad datasheet	INGESTED + precision extract
ATEX already held; remaining cert path	In roadmap Sec.7
Altium Designer course (Telemetry)	INGESTED altium-course-*
OrCAD PSpice (Telemetry)	INGESTED orcad-pspice-manual
Electronics handbooks (Telemetry)	INGESTED Newnes / Bowick / Jung / "I
Pin budget / interface list	MISSING - next deliverable
Lab scoping quotes (62443/61850)	MISSING
Variscite / Compulab kit scorecards	MISSING - fill Table B in vendor selection doc

2.0.3 Mapping summary (quick view)

KB branch	Ingested now?	Priority to fill
Yocto Docs	Yes - 5.0.13 mega-manual + BSP notes	MEDIUM - vendor SoM BSP after SKU freeze
BitBake Docs	Yes - 2.18 extract	LOW - build runbook after first image
Custom Layers	No	HIGH at first BSP bring-up
Recipes (.bb)	No	MEDIUM-HIGH with apps
Kernel & Device Trees	SoC DS only	HIGH with carrier schematic
Application Source Code	Design docs only	HIGH for Phase 1 firmware
Protocol Documentation	CEI O/T, regs class, EU RED/CRA, ref manuals	HIGH - ARERA, 62351, 62443, 61850 PDFs
Engineering Knowledge Base	Strong start	Continue (pin budget, cert quotes)

2.0.4 Recommended folder layout (on disk -> RAG)

```
knowledge-base/  
âââââ README.md                                # this brief (= CCI_Knowledge_Base_Brief.md)  
âââââ DOWNLOADS.md                              # MASTER path <-> URL <-> source_id index  
âââââ 01-yocto/  
â ,      âââââ yocto-5.0.13-theyoctoproject.pdf          -> ccli-yocto-overview  
â ,      âââââ NXP_UG10164_iMX_Yocto_Reference.md         -> ccli-nxp-yocto-ug  
â ,      âââââ CCI_GitHub_Yocto_References.md             -> ccli-github-yocto-refs  
â ,      âââââ github-refs/*.md  
â ,      âââââ DOWNLOADS.md  
âââââ 02-bitbake/  
â ,      âââââ bitbake-user-manual-scarthgap-era-extract.md -> ccli-bitbake-manual
```

```

â"œâ"€â"€ 03-custom-layers/ "| 06-application/
â"œâ"€â"€ 07-protocols/
â",   â"œâ"€â"€ CCI_Module_Regulations_Classification.md -> ccli-cci-module-regs-class
â",   â"œâ"€â"€ CELEX_32018L2001_EN_TXT.pdf             -> eu-2018-2001-red
â",   â""â"€â"€ OJ_L_202402847_EN_TXT.pdf               -> eu-2024-2847-cra
â""â"€â"€ 08-engineering/
    â"œâ"€â"€ CCI_Vendor_Selection.md                   -> ccli-vendor-selection
    â"œâ"€â"€ Toradex_Verdin_Development_Board_V1.1_Datasheet.pdf -> ccli-toradex-verdin-devbo
    â"œâ"€â"€ Toradex_Verdin_Carrier_Board_Design_Guide.pdf   -> ccli-toradex-carrier-guid
    â""â"€â"€ Toradex_Verdin_Family_Specification.pdf       -> ccli-toradex-verdin-famil

```

Repo root (also ingested via Telemetry projects/ccli/):

```

â"œâ"€â"€ CCI_Project_Roadmap.md           -> ccli-project-roadmap
â"œâ"€â"€ CCI_SoM_Comparison.pdf           -> ccli-som-comparison
â"œâ"€â"€ CCI_Clone_RE_Option.pdf          -> ccli-clone-re-option
â"œâ"€â"€ CCI_Yocto_BSP_Notes.md           -> ccli-yocto-bsp-notes
â""â"€â"€ CCI_Knowledge_Base_Brief.md      # sync copy of knowledge-base/README.md

```

Telemetry regulations (not under knowledge-base/):

```

â"œâ"€â"€ regulations/cei-0-16/"|
â""â"€â"€ regulations/eu/"|               -> eu-2018-2001-red, eu-2024-2847-cra

```

Ingest convention: `source_id = ccli-kb-<branch>-<shortname>` for new KB-only docs Example: `ccli-kb-yocto-vendor-bsp`, `ccli-kb-dt-carrier`, `ccli-kb-app-iec61850`.

2.0.5 To do

1. **Lock SoM SKU** (prefer Verdin/DART Plus IT) + order **eval kit** (Toradex **99991105** lead path) - see `ccli-vendor-selection`
2. **Vendor Yocto/BSP PDF** for chosen SoM -> `ccli-yocto-vendor-bsp`, kernel/DT baseline
3. ~Drop official UG10164 PDF~~ **done** - re-ingest with `ingest-ccli.ps1`
4. **ARERA 540/2021 + CEI EN 61557-12** -> Italian mandate + metrology
5. **IEC 62443-4-1/4-2 + IEC 62351-3**|6 -> cyber certification corpus
6. **Connector / pin-budget (Altium)** -> kernel/DT + engineering KB
7. **Energy analyzer Modbus map** -> protocol documentation
8. **IEC 61850 stack decision + ICD outline** (libiec61850 GPLv3 + MZ commercial to ship) -> application + protocols
9. **meta-ccli README** (skeleton OK) -> custom layers + recipes index

2.0.6 Telemetry RAG source_ids already useful for this KB

source_id	KB branch
ccli-knowledge-base-brief	This document (tree + status)
ccli-yocto-overview	Yocto Docs - 5.0.13 Scarthgap mega-manual
ccli-bitbake-manual	BitBake Docs - 2.18 extract
ccli-nxp-yocto-ug	NXP UG10164 official PDF (LF6.18.20)
ccli-github-yocto-refs	imx-manifest, meta-freescale, libiec61850, meta-secure-core, "
ccli-yocto-bsp-notes	Yocto Docs (gap + plan)
ccli-toradex-verdin-devboard / -carrier-guide / -verdin-family	Toradex kit PDFs
ccli-cci-module-regs-class	Protocol / regulations classification
cei-0-16-allegato-o / -t	Protocol Documentation
cei-0-16-cci-manual-ailux	Protocol / reference (non-clone)
eu-2018-2001-red / eu-2024-2847-cra	Protocol / EU cyber & energy
ccli-project-roadmap	Engineering KB
ccli-som-comparison	Engineering KB / Kernel platform
ccli-vendor-selection	Engineering KB - vendor/kit tables (Toradex 99991105)
ccli-nxp-imx8m-dqlq-cec (+ -precise)	Kernel platform / SoC
ccli-clone-re-option	Engineering KB (Option B)
ccli-igate-850-manual / ccli-moxa-mgate-5119	Protocol / gateways
altium-course-*	Engineering KB (EDA)
orcad-pspice-manual	Engineering KB (simulation)
Newnes / Bowick / Jung / "	Engineering KB (electronics theory)

2.0.7 Keywords (routing)

knowledge base, yocto, bitbake, meta-ccli, recipe, .bb, .bbappend, device tree, dts, kernel, iec 61850, modbus, pf2, hab, secure boot, certified path, ccli-kb

II Vendor selection & prototype BOM

Chapter 3

Vendor selection (C1-C10)

3.1 CCI Vendor Selection - Product Analysis & Dev-Kit Matrix

Document ID: CCLI-VENDOR-SEL-001 **Revision:** 1.1 **Date:** 2026-07-25 **RAG source_id:** ccli-vendor-selection **Purpose:** Analyze each SoM vendor's **product** (SoM + eval kit + optional industrial gateway) against the CEI 0-16 / AiLux-class CCI control set, then choose a **development kit** for Phase 1 firmware - without treating the kit as the field product. **Companions:** CCI_SoM_Comparison (silicon), CCI_Project_Roadmap (Option A), CCI_Module_Regulations_Classification (protocol/cyber). **Download index:** ../DOWNLOADS.md

3.1.1 Decision rule (read first)

Layer	Role	Deploy as CCI?
SoM	Production compute on custom carrier	Yes (with carrier)
Dev / evaluation board	BSP, Linux, early 61850/Modbus, signed-boot experiments	No
Industrial gateway (vendor SKU)	Reference for I/O patterns / BSP maturity	Usually No - form factor & CEI Eth_A/B story rarely match

Phase 1 buy: one vendor's SoM + matching eval kit. **Phase 1 build:** custom carrier for missing controls (extra Ethernet, isolated RS-485, field DI/DO, GNSS/LTE).

3.1.2 Table A - CCI control checklist (master requirements)

ID	Requirement	Prototype kit	Product (carrier + SoM)
C1	$\geq 4 \times 10/100$ Ethernet: Eth_A (DSO), Eth_B (Operatore Abilitato / Qualified Operator), $2 \times$ plant	Partial OK (2 ports for SW demo)	Required
C2	$2 \times$ opto-isolated RS-485 (Modbus RTU) on RJ45-class plant serial	$1 \times$ UART/RS232 OK for Modbus demo	Required
C3	IEC 61850 Server/Client/GOOSE on Eth_A	Required (SW on kit)	Required
C4	DI $10-120$ Vdc $\times 5$ (external field supply) / DO relay $\times 3$	Stub in SW OK	Required
C5	GNSS (GPS/GLONASS) time sync + SMA path	Nice-to-have	Required
C6	Secure boot + key storage (HAB + SE/TPM path)	BSP proof on kit	Required
C7	Maintained Yocto/Torizon BSP, industrial-temp SoM SKU	Required	Required
C8	Longevity / PCN (~ 10 -year industrial)	No	Required
C9	Vendor signed-boot / security docs	Strongly preferred	Required
C10	Power $12-24$ Vdc class, DIN-rail product path	Eval may differ	Required

Scoring keys used below: **Full** / **Partial** / **None** / **N/A**.

3.1.3 Table B - Vendor product shortlist (analyze each company)

Vendor	SoM SKU (shortlist)	Dev / eval kit	Industrial reference	BSP / reference	Secure-boot path	Kit est. €
Toradex	Verdin iMX8M Plus IT (preferred) or Mini IT	Verdin Development Board with HDMI Adapter - PN 99991105 (see Sec.3)	- (custom carrier)	Torizon / Yocto; [toradex-manifest](https://github.com/toradex-manifest.git)	[build-signed-imx8m-bootloader](https://github.com/toradex/build-signed-imx8m-bootloader) [meta-toradex-security](https://github.com/toradex-meta-toradex-security)	Board ~€300 + SoM
Variscite	DART-MX8M-PLUS IT (or MINI)	Vendor DART carrier / eval	VAR-SOM / SMARC MX8M-PLUS family	[variscite-bsp-platform](https://github.com/varigit/variscite-bsp-platform), [meta-variscite-imx](https://github.com/varigit/meta-variscite-imx) (Scarthgap 6.6.y)	Vendor HAB docs	TBD
Compulab	CL-SOM-iMX8 / MCM-iMX8M-Mini	Compulab eval carrier	IOT-GATE-iMX8 industrial gateway	[meta-bsp-imx8mm](https://github.com/compulab-yokneam/meta-bsp-imx8mm)	TBD	TBD
NXP	i.MX 8M Plus EVK SoC path	EVK-MIMX8MP	Reference only	[imx-manifest](https://github.com/nxp-imx/imx-manifest), UG10164	HAB / CAAM reference [nxp](https://github.com/nxp)	Higher

Base layers (all vendors): [meta-freescale](https://github.com/Freescale/meta-freescale) (scarthgap = Yocto 5.0 LTS), [meta-imx](https://github.com/nxp-imx/meta-imx) for NXP-release recipes, [meta-secure-core](https://github.com/jiazhang0/meta-secure-core) / [meta-security](https://git.yoctoproject.org/meta-security) for TPM/hardening.

3.1.4 Toradex kit row - Verdin Development Board (PN 99991105)

3.1 Product card (from vendor listing)

Field	Value
Name	Verdin Development Board with HDMI Adapter
PN	99991105
Version	V1.1F
Status	Sample Production
Intended use	Evaluation / Development (not field CCI)
Unit price (listing)	€300.00
Supply	7-24 V DC $\pm 10\%$
Dimensions	250 mm x 200 mm
USB	2x USB-A 3.0 Host; 1x USB-C 2.0 DualRole
Ethernet	2x Gigabit Ethernet
Serial / bus	1x RS232; 2x CAN; 4x I2C; 1x SPI
I/O	3x PWM; 4x ADC; up to 98x GPIO
Audio / video	Stereo in/out; HDMI adapter

SoM to pair (recommended): Verdin **iMX8M Plus** Industrial-temp SKU (dual GbE MAC on SoC - cleaner Eth_A / Eth_B story than Mini). Mini IT remains a cost-down option with more carrier Ethernet work.

3.2 Table - Toradex kit vs CCI controls (C1-C10)

ID	CCI need	Kit / SoM score	Notes
C1	4x Ethernet Eth_A/B + plant	Partial	Board has 2x GbE only - enough for early Eth_A + Eth_B SW demo; 2 plant ports + switch/PHY on custom carrier
C2	2x isolated RS-485	Partial	Board has 1x RS485 (DB9 via UART1 transceiver) + 1x RS232 - not opto-isolated and not 2x RJ45 Modbus; product C2 still on custom carrier
C3	IEC 61850 on Eth_A	Full (SW)	Run libiec61850 / commercial stack on Torizon/Yocto over one GbE port
C4	DI 10-120 Vdc / DO relays	None	GPIO only - no field-voltage conditioning; stub DO in SW until carrier
C5	GNSS	None on board	Add u-blox-class module on carrier (UART/I2C + SMA)
C6	Secure boot / TPM path	Partial -> Full	SoM HAB + [signed bootloader container](https://github.com/toradex/build-signed-imx8m-bootloader); optional TPM on Plus SKUs - verify SKU
C7	Yocto / industrial SKU	Full	Mature Toradex BSP / Torizon; IT SoM SKUs available
C8	Longevity	N/A (kit) / Full (SoM IT)	Request 10-year / PCN letter for chosen Verdin SKU
C9	Signed-boot docs	Full	meta-toradex-security + signed-boot GitHub project
C10	DIN-rail product power/form	Partial	7-24 V OK for bench; not DIN-rail CCI enclosure

3.3 What this kit is for vs not for

Use now (Phase 1)	Defer to custom carrier
Boot Linux / Torizon, pin BSP revision	Extra 2x Ethernet + managed switch / PHYs
Dual GbE networking, VLAN experiments	2x opto-isolated RS-485 Modbus
libiec61850 MMS/GOOSE prototype on Eth_A	Field DI 10-120 Vdc + relay DO
Secure-boot / signed image chain trials	GNSS + LTE modules, anti-tamper, OLED
Local USB console / diagnostics	DIN-rail mechanical + product EMC

3.1.5 Table C - SoM silicon shortlist (summary; detail in SoM comparison)

Option	SoC	On-chip GbE	RT core	Ind. temp	CCI note
Toradex Verdin iMX8M Mini	8M Mini	1 (+TSN)	M4	Yes (IT)	Lower cost; harder Eth_A/B isolation
Toradex Verdin iMX8M Plus	8M Plus	2	M7	Yes (IT)	Preferred with kit 99991105
Variscite DART-MX8M-MINI	8M Mini	1	M4	Yes	Pin2pin DART
Variscite DART-MX8M-PLUS	8M Plus	2	M7	Yes	Strong Toradex alternative
Compulab CL-SOM-8M class iMX8	8M class	1 typ.	M4	Check	Confirm SKU + longevity
NXP EVK / bare SoC	8M Dual/Quad/Plus	1-2	M4/M7	Check	Reference, not DIN SoM

See also CCI_SoM_Comparison certificate map (62443 / 61850 / 62351 vs silicon).

3.1.6 Table D - Phase 1 kit recommendation

Goal	Recommended buy	Validates	Still on carrier
Lead path (recommended)	Toradex Verdin Dev Board 99991105 + Verdin iMX8M Plus IT SoM	C3, C6-C7, C9; Partial C1 (2x Eth)	C1 remainder, C2, C4, C5, C10 product form
Cost-down eval	Same board + Verdin Mini IT	Same SW path; weaker Ethernet	Same + harder Eth split
Parallel reference	Compulab IOT-GATE-iMX8 (study only)	Industrial I/O patterns	Not CEI CCI form factor
Silicon/HAB only	NXP EVK-MIMX8MP	HAB, DT, UG10164	Everything else

Architecture Freeze gate: (1) SoM SKU locked, (2) one eval kit ordered, (3) carrier block diagram frozen for C1/C2/C4/C5.

3.1.7 Scorecard template (fill when datasheets arrive)

Copy one block per vendor product (SoM / kit / gateway).

Vendor / PN	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	Evidence (PDF/URL)
Toradex 99991105 (+ Plus SoM)	P	N	F*	N	N	P/F	F	SoM	F	P	This doc Sec.3
Variscite											
...											
Compulab											
...											
NXP EVK ...											

*F for software capability on kit Ethernet, not full 4-port plant topology.

3.1.8 Toradex download paths (local + URL)

Artifact	Local path (under knowledge-base/)	URL
Dev Board V1.1 datasheet	08-engineering/Toradex_Verdin_Development_Board_V1.1_Datasheet.pdf	https://www.toradex.com/en/109262-verdin-development-board-datasheet.pdf
Carrier board design guide	08-engineering/Toradex_Verdin_Carrier_Board_Design_Guide.pdf	https://www.toradex.com/en/108140-verdin-carrier-board-design-guide.pdf
Verdin family specification	08-engineering/Toradex_Verdin_Family_Specification.pdf	https://www.toradex.com/en/109262-verdin-family-specification.pdf
Kit product page (99991105)	-	https://www.toradex.com/products/carrier-board/verdin-development-board-kit
Developer center	-	https://developer.toradex.com/hardware/verdin-som-family/carrier-boards/verdin-development-board
Signed bootloader (GitHub)	01-yocto/github-refs/toradex-signed-bootloader	https://github.com/toradex/build-signed-imx8m-bootloader

RAG: ccli-toradex-verdin-devboard, ccli-toradex-carrier-guide, ccli-toradex-verdin-family.

3.1.9 To do

1. Confirm exact **Verdin iMX8M Plus IT** orderable SKU (RAM/eMMC/Wi-Fi - prefer **no Wi-Fi** for CCI); download SoM datasheet into 08-engineering/.
2. Request Toradex **longevity** / **PCN** letter for that SKU.
3. Fill Table B rows for Variscite and Compulab with PNs and prices.
4. Keep **libiec61850** as prototype stack (**GPLv3**; MZ Automation **commercial license to ship**) - see roadmap Sec.5.
5. Drop official **UG10164 PDF** at 01-yocto/NXP_UG10164_iMX_Yocto_Project_Users_Guide.pdf (see ./DOWNLOADS.md).

3.1.10 Keywords (RAG)

vendor selection, toradex, verdin, 99991105, development board, dart, variscite, compulab, iot-gate, eval kit, eth_a, eth_b, carrier, som shortlist, ccli-vendor-selection, download path, 116796

Chapter 4

Vendor decision scorecard

4.1 CCI Vendor Decision Scorecard (Phase 1) - post Wave 1 ingest

Website Chapter 07 · `website/src/lib/vendorScorecard.ts` + `validationGate.ts`. **Rule:** Validate datasheet evidence **before** treating scores as a buy decision.

4.1.1 Wave 1 ingest (2026-07-25)

source_id	Status
ccli-toradex-verdin-plus-som	completed
ccli-variscite-dart-mx8m-plus	completed
ccli-variscite-dt8m-customboard	completed
ccli-compulab-iot-gate-ref	completed
ccli-vendor-decision-scorecard	completed

Still **MISSING** for silicon industrial claims: `IMX8MPIEC.pdf` (manual NXP download).

4.1.2 Evidence gates

Candidate	Gate	Meaning
Toradex Verdin 99991105 + 0063	CONDITIONAL	SoM + kit DS ingested - OK to order; PCN still open
Variscite DART + DT8MCustomBoard	PARTIAL	Dual GbE DS ingested - lock IT PN before buy
NXP 8MPLUSLPD4-EVK	SILICON ONLY	IEC DS still missing
Compulab CL-SOM EVK	BLOCKED	No PDFs in KB
Compulab IOT-GATE	STUDY ONLY	Ref guide ingested - not CEI CCI

4.1.3 Ranking (after Wave 1)

Rank	Candidate	Score	Role
1	Toradex Verdin Dev 99991105 + 84 Plus IT 0063		Lead
2	Variscite DART-MX8M-PLUS kit 67 path		Alternate
3	NXP 8MPLUSLPD4-EVK 60		Silicon / HAB
4	Compulab CL-SOM EVK 42		Blocked
5	Compulab IOT-GATE-iMX8 41		I/O study only

Chapter 5

Vendor kit catalog

5.1 CCI Vendor Kit Catalog (Prototype BOM)

Source of truth for website Chapter 06 · Prototype BOM ([website/src/lib/bomCatalog.ts](#)). **Rule:** Validate official datasheets first, then score kits against C1-C10. Eval kits are never the field CCI.

5.1.1 Lead pick (Phase 1)

Item	Full name	PN
Eval kit	Verdin Development Board with HDMI Adapter	99991105
SoM	Verdin iMX8M Plus Quad 4GB IT	0063

5.1.2 Supplier categories

Toradex (lead)

- Verdin Development Board with HDMI Adapter - **99991105**
- Verdin iMX8M Plus Quad 4GB IT - **0063**
- Verdin iMX8M Plus Quad 4GB Wi-Fi / Bluetooth IT - **0058** (radio usually unwanted)
- Dahlia Carrier Board kit (compact; typically 1x GbE)
- Verdin iMX8M Plus Evaluation Kit 1 (shop bundle - confirm SoM SKU)

Variscite (alternate - validate before compare)

- VAR-DVK-DT8M-PLUS / VAR-STK-DT8M-PLUS + DART-MX8M-PLUS
- VAR-SOM-MX8M-PLUS Evaluation Kit (Symphony-Board V2)
- SoMs: DART-MX8M-PLUS, VAR-SOM-MX8M-PLUS

Compulab

- CL-SOM-iMX8 and SBC-iMX8 Evaluation Kit (alternate SoM)
- IOT-GATE-iMX8 Evaluation Kit / IOT-GATE-iMX8 - **I/O study only**, not CEI Eth_A/B CCI

NXP (silicon / HAB)

- **8MPLUSLPD4-EVK** - silicon / HAB / Yocto lab
- IMX8MPCEC - consumer datasheet (on disk)
- IMX8MPIEC - industrial datasheet (**MISSING** - required for IT claims)

5.1.3 Custom carrier (after SoM lock)

Shared lines C1 Ethernet · C2 RS-485 · C4 DI/DO · C5 GNSS · C6 security · C10 power - see CCI_Prototype_BOM.md.

Chapter 6

Prototype BOM & missing index

6.1 CCI Prototype BOM & Missing Document Index

Document ID: CCLI-BOM-001 **Revision:** 1.1 **Date:** 2026-07-25 **RAG** **source_id:** ccli-prototype-bom
Companion JSON: CCI_Prototype_BOM.json **Budget target:** ~€1,000 single-unit Phase 1 (see CCI_Project_Roadmap Sec.3) **Lead path:** Toradex Verdin iMX8M Plus Quad 4GB IT (**PN 0063**) + custom carrier **Lead eval kit:** Verdin Development Board with HDMI Adapter (**PN 99991105**) - SW / BSP only **Purpose:** Single place to **store the prototype BOM**, list **every datasheet already downloaded**, and mark what is **still MISSING** before schematic / Architecture Freeze. Status keys: **HAVE** (on disk) · **INGESTED** (on disk + Telemetry RAG) · **PARTIAL** · **MISSING** · **TBD** (part not selected yet). **Companions:** CCI_Vendor_Selection.md · CCI_Vendor_Kit_Catalog.md · CCI_Vendor_Decision_Scorecard.md · CCI_Mocking_Bench_BOM.md (lab CapEx) · [../DOWNLOADS.md](../DOW

6.1.1 Snapshot - downloaded vs still missing

0.1 Datasheets & engineering PDFs downloaded (on disk)

File	Role	source_id	RAG
Toradex_Verdin_iMX8M_SoC_HW_Datasheet_0063_Quad_4GB_IT_no_Wi-Fi.pdf	SoC HW Datasheet	ccli-toradex-verdin-plus-4gb	INGESTED
Toradex_Verdin_Development_Kit_99991105_Datasheet.pdf	Development Kit Datasheet	ccli-toradex-verdin-dev-kit	INGESTED
Toradex_Verdin_Carrier_Board_Design_Guide.pdf	Carrier Board Design Guide	ccli-toradex-carrier-board	INGESTED
Toradex_Verdin_Family_Visual_Specification.pdf	Visual Specification	ccli-toradex-verdin-family	INGESTED
reference/IMX8MPCEC.pptx	NXP i.MX 8M Plus SoC (CEC)	ccli-nxp-imx8m-plus-cec	INGESTED
reference/IMX8MDQLQCEC-1280MX-SM.pdf	NXP i.MX 8M Dual/Quad Lite SoC	ccli-nxp-imx8m-dqlq-cec	INGESTED
../01-yocto/NXP_UG10184_NXP_Yocto_HAB_project_guides_Guide.pdf	NXP Yocto HAB project guides	ccli-yocto-ug	INGESTED
reference/vendor-validation/Verdin-Kit-DART-iMX8M-PLUS_Datasheet.pdf	Verdin-Kit-DART-iMX8M-PLUS Datasheet	ccli-dart-mk8m-plus	INGESTED
reference/vendor-validation/Verdin-Kit-DART-iMX8M-PLUS_CustomBoard_Datasheet.pdf	Verdin-Kit-DART-iMX8M-PLUS CustomBoard Datasheet	ccli-dart-mk8m-plus-custom-board	INGESTED
reference/vendor-validation/Verdin-Kit-DART-iMX8M-PLUS_ReferenceModule.pdf	Verdin-Kit-DART-iMX8M-PLUS ReferenceModule	ccli-dart-mk8m-plus-ref-module	INGESTED
reference/moxa-mgate-5119-segates-datasheet-ref.1.2.pdf	Moxa iGate-850 V101UMEN gateway pattern ref	ccli-moxa-mgate-5119	INGESTED
reference/iGate-850_V101UMEN.pdf	Moxa iGate-850 V101UMEN gateway pattern ref	ccli-igate-850-manual	INGESTED
reference/STM32L151RCT-Companion MCU pattern ref	STM32L151RCT Companion MCU pattern ref	ccli-stm32l151rct	INGESTED

0.2 Authored engineering docs HAVE

File	source_id	Notes
CCI_Vendor_Selection.md	ccli-vendor-selection	C1-C10 kit vs carrier
CCI_Vendor_Kit_Catalog.md	-	Kit full names / PNs
CCI_Vendor_Decision_Scorecard.md	ccli-vendor-decision-scorecard	Scores after Wave 1
CCI_Industrial_RAG_Architecture.md	ccli-industrial-rag-arch	Hybrid RAG design
CCI_Validation_and_Mocking_Strategy.md	ccli-validation-strategy	L1-L4 PF2 bench ladder
CCI_Mocking_Bench_BOM.md	ccli-mocking-bench-bom	Lab CapEx Waves A-D
cci-knowledge-graph.json	-	C1-C10 vs kit edges

0.3 Still MISSING (must collect or author)

Priority	Item	Target path / source_id	Blocks
P0	NXP IMX8M+IEC (industrial SoC DS)	reference/IMX8M+IEC.pdf -> ccli-nxp-imx8m-plus-iec	Industrial silicon claims
P0	Toradex PCN / longevity letter (SKU 0063)	vendor PDF	C8 / Architecture Freeze
P0	Pin budget / connector map	CCI_Pin_Budget.md -> ccli-pin-budget	Schematic freeze
P0	Carrier block diagram (frozen)	CCI_Carrier_Block_Diagram.md -> ccli-carrier-block	C1/C2/C4/C5 freeze
P1	Ethernet switch / PHY + mag-jack DS	reference/<MPN>-...	Carrier C1 (MPN TBD)
P1	2x opto-isolated RS-485 transceiver DS	reference/<MPN>-...	Carrier C2 (MPN TBD)
P1	DI opto + DO relay datasheets	reference/<MPN>-...	Carrier C4 (MPN TBD)
P1	GNSS module + antenna DS	reference/<MPN>-...	Carrier C5 (MPN TBD)
P1	Power tree (12-24 Vdc -> SoM rails) DS	reference/<MPN>-...	Carrier C10 (MPN TBD)
P1	Energy analyzer Modbus register map	07-protocols/... -> ccli-modbus-analyzer-map	Plant serial software
P2	Discrete TPM DS (if not on SoM assembly)	reference/<MPN>-...	C6 product claim
P2	Compulab CL-SOM-iMX8 eval kit PDFs	reference/vendor-validation/	Unlock Compulab SoM score
P2	Variscite Symphony / VAR-SOM-Plus pack	optional	Second Variscite path
P2	DIN-rail enclosure drawing	mechanical	C10 product form
P2	Altium ActiveBOM export	after schematic	Fab BOM
P3	Anti-tamper switch + accelerometer DS	product	CRA / 62443 story
P3	OLED / HMI DS	optional proto	Local diagnostics
P3	LTE Cat.1 module DS (CCI MPN)	optional rev A	PARTIAL generic refs only
P3	ARERA 540/2021, CEI EN 61557-12, IEC 62443, IEC 62351	licensed	Product cert track - not kit pick

6.1.2 Document gate (must exist before schematic freeze)

Clustered PDF summary: see CCI_RAG_Knowledge_Base.md (Part VI). Full line-item detail below.

#	Document	Status	Target path / source_id	Why
D0	Architecture block diagram	HAVE	CCI_Architecture.png / mermaid	Functional interfaces
D1	Project roadmap + BOM cost envelope	HAVE	ccli-project-roadmap	Budget / phases
D2	Vendor selection + C1-C10 scorecard	HAVE	ccli-vendor-selection	Kitors carrier split
D3	Pin budget / connector map	MISSING	CCI_Pin_Budget.mxd -> ccli-pin-budget	Every SoM ball -> net -> connector
D4	Carrier block diagram (frozen)	MISSING	CCI_Carrier_Block_Diagram.mxd -> ccli-carrier-block	C1/C2/G4/G5 freeze gate
D5	Verdin Plus IT SoM datasheet	HAVE / INGESTED	Toradex_Verdin_iMX8MPPlus_datasheet.pdf -> ccli-toradex-verdin-plus-som	DSM0063-17-01 in BOM
D5a	NXP i.MX 8M Plus CEC SoC DS	HAVE / INGESTED	reference/IMX8MPCSoC.pdf	SiCipref
D5b	NXP i.MX 8M Plus IEC SoC DS	MISSING	reference/IMX8MPCIEC.pdf -> ccli-nxp-imx8m-plus-iec	IEC official-temp silicon claims
D6	Longevity / PCN letter (SKU 0063)	MISSING	vendor PDF	10-year industrial path
D7	ActiveBOM export (Altium)	MISSING	after schematic	Fab-ready MPN list
D8	Energy analyzer Modbus register map	MISSING	07-protocols/... -> ccli-modbus-analyzer-map	RS-485 plant serial
D9	GNSS module datasheet (chosen MPN)	MISSING	reference/	UART/I2C + SMA + PPS
D10	LTE module datasheet (if rev A)	PARTIAL	Quectel-class refs in Telemetry; CCI MPN TBD	Optional on rev A
D11	Relay + opto datasheets (DI/DO)	MISSING	per MPN	Field 10-120 Vdc / 250 Vac DO
D12	Ethernet switch / PHY datasheets	MISSING	per MPN	2 plant ports beyond SoM GbE
D13	Isolated RS-485 transceiver DS	MISSING	per MPN	2x plant Modbus
D14	Power (12-24 Vdc -> SoM rails) DS	MISSING	buck/LDO/PoE TBD	C10 product path
D15	TPM / secure element DS (if discrete)	MISSING	if not on SoM SKU	C6 key storage
D16	Anti-tamper switch + accelerometer DS	MISSING	product; optional proto	CRA / 62443 story
D17	OLED / HMI DS (optional proto)	MISSING	optional	Local diagnostics
D18	Enclosure / DIN-rail mechanical	MISSING	drawing + BOM	Not eval board form
D19	EMC / connector family (RJ45 8/8)	PARTIAL	CEI manual behaviour HAVE ; part PNs TBD	Plant serial pinout
D20	Variscite DART + DT8M board (alternate)	HAVE / INGESTED	reference/vendor-Dart-GbE	Dart GbE/alternate path
D21	Compulab IOT-GATE	HAVE / IN-	reference/vendor-IoT-gate	IoT gate only

6.1.3 Architecture & system (non-PCB but BOM-driving)

Item	Qty	Est. €	Doc status	Notes
Functional architecture (Eth_A/B, plant, GNSS, DI/DO)	-	-	HAVE	Diagram + CEI O/T
Regulations classification (feature -> norm)	-	-	HAVE	ccli-cci-module-regs-class
Industrial RAG graph (C1-C10 vs kit)	-	-	HAVE	cci-knowledge-graph.json
Vendor decision scorecard (Wave 1)	-	-	HAVE	Toradex 84 · Variscite 67
Architecture Freeze checklist	-	-	PARTIAL	SoM DS HAVE ; pin budget + PCN still open

6.1.4 Compute - SoM + eval (Phase 1 buy)

Ref	Description	Locked / suggested	Qty	Est. €	Datasheet / doc	Status
U-SOM	Verdin iMX8M Plus Industrial SoM	0063 Quad 4GB IT (no Wi-Fi)	1	150-350	SoM DS + family spec	HAVE ccli-toradex-verdin-plus-som
KIT	Verdin Development Board + HDMI	PN 99991105	1	~300	Dev board DS	HAVE ccli-toradex-verdin-devboard
-	Carrier design guide	-	-	-	Toradex carrier guide	HAVE ccli-toradex-carrier-guide
-	Verdin family specification	-	-	-	Family PDF	HAVE ccli-toradex-verdin-family
-	NXP i.MX 8M Plus SoC	Silicon / Yocto ref	-	-	CEC + IEC + Dual-Lite	CEC HAVE ; DualLite HAVE ; IEC MISSING
ALT	Variscite DART-MX8M-PLUS + DT8M board	Alternate (IT PN TBD)	-	-	DART + board DS	HAVE (compare only)
STUDY	Compulab IOT-GATE-iMX8	I/O patterns only	-	-	Reference guide	HAVE (not deploy as CCI)

Kit validated facts (Dev Board DS): 2x GbE · 1x RS485 (DB9, **not** opto) · 1x RS232 · isolated CAN. Product still needs carrier for full C1/C2/C4/C5/C10. **BOM note:** Kit is **not** the field product. Carrier replaces kit I/O for C1 remainder, full C2, C4, C5, C10.

6.1.5 Ethernet & plant networking (C1)

Ref	Description	Qty	Est. €	Doc needed	Status
SW1	4-port 10/100 (or GbE) switch 1 IC or 2x PHY + SoM dual MAC		8-20	Switch/PHY DS + magnetics	MISSING (MPN TBD)
PHY	Additional Ethernet PHY for plant ports	0-2	3-8	PHY DS	MISSING
J-ETH	RJ45 8/8 + magnetics (Eth_A, Eth_B, Plantx2)	4	4-10	Connector / magjack DS	MISSING
-	Isolation / VLAN policy note	-	-	Carrier Eth_A/B de- sign note	MISSING

Have for behaviour: CEI Annex O/T, AiLux manual (port roles) - not part PNs. **Have for lab Eth_A/B SW:** Toradex kit 2x GbE + Variscite DART 2x GbE (eval only).

6.1.6 Serial / Modbus plant (C2) - signals

Ref	Description	Qty	Est. €	Doc needed	Status
U-485	Opto-isolated RS-485 transceiver	2	10-15	Transceiver DS	MISSING (MPN TBD)
J-SER	RJ45 8/8 Modbus pinout plant serial	2	2-4	Pinout map + connector DS	MISSING (pinout from CEI/manual PARTIAL)
-	Energy analyzer Modbus map	-	-	Register map PDF	MISSING ccli-modbus-analyzer-map
-	Reference gateways (patterns only)	-	-	Moxa / iGate manuals	HAVE (not BOM items)

Kit note: Toradex board has 1x RS485 DB9 (lab Partial) - **not** CEI 2x opto RJ45.

6.1.7 Digital I/O - DI signals & DO relays (C4)

Ref	Description	Qty	Est. €	Doc needed	Status
U-DI	Opto-isolator / DI conditioner 5 ch (10-120 Vdc, external field supply)		8-15	Opto / DI mod- ule DS	MISSING (MPN TBD)
K-DO	Relay DO (dry contact / 250 Vac 3 A class)	3	5-12	Relay DS + footprint	MISSING (MPN TBD)
-	Flyback diodes, snubbers, TVS	lot	2-5	Discrete DS	MISSING
J-IO	Field terminal blocks (DI/DO) 1 set		3-8	Terminal DS	MISSING
-	PF2 curtailment signal defini- tion	-	-	App note / ICD	MISSING

6.1.8 Time sync - GNSS (C5)

Ref	Description	Qty	Est. €	Doc needed	Status
U-GNSS	GPS/GLONASS module (u- blox-class)	1	20-35	Module DS	MISSING (MPN TBD)
ANT-G	Active GNSS antenna + SMA- F path	1	5-15	Antenna DS	MISSING
-	PPS / time sync note (IEEE 1588 optional)	-	-	App note	MISSING

6.1.9 Cellular - LTE (optional rev A)

Ref	Description	Qty	Est. €	Doc needed	Status
U-LTE	LTE Cat.1 (Quectel-class)	0-1	25-40	Module HW design guide	PARTIAL (generic refs; CCI MPN TBD)
ANT-L	LTE antenna SMA	0-1	5-10	Antenna DS	MISSING
SIM	Nano-SIM holder	0-1	1-2	Holder DS	MISSING

6.1.10 Security / trust (C6)

Ref	Description	Qty	Est. €	Doc needed	Status
-	HAB / secure boot (on SoM)	-	-	UG10164 + Toradex SoM DS (HAB Yes) + signed boot	HAVE
U-TPM	Discrete TPM 2.0 (if not on SoM SKU)	0-1	5-15	TPM DS + FIPS cert note	MISSING until assembly option freeze
-	Secure element (optional)	0-1	TBD	SE DS	TBD
SW-AT	Anti-tamper cover switch	0-1	1-3	Switch DS	MISSING (product)
U-ACC	XYZ accelerometer anti-tamper	0-1	2-5	Accel DS	MISSING (product)

6.1.11 Power (C10)

Ref	Description	Qty	Est. €	Doc needed	Status
U-PWR	12-24 Vdc input -> SoM + 3.3/1.8 rails	1	10-25	Buck/LDO/hot-swap DS	MISSING (MPN TBD)
F1	Input fuse / eFuse / reverse polarity	1	1-3	Protection DS	MISSING
-	Field DI external supply isolation note	-	-	Design note	MISSING

Kit note: Dev Board supply 7-24 V OK for bench - not DIN-rail CCI enclosure.

6.1.12 Mechanical / HMI / misc

Ref	Description	Qty	Est. €	Doc needed	Status
PCB	4-layer carrier, proto qty 5-10	1 lot	50-150	Gerbers (later)	MISSING
ENC	DIN-rail enclosure	1	15-40	Mechanical drawing	MISSING
DISP	OLED 1" (optional)	0-1	5-15	Display DS	MISSING
USD	MicroSD holder	0-1	1-2	Holder DS	TBD
-	Fab BOM (Altium Active-BOM)	-	-	CSV/PDF export	MISSING

6.1.13 Software / stack (not PCB BOM, but prototype “bill”)

Item	Status	Doc / license note
Yocto 5.0.13 / BitBake	HAVE	manuals ingested
Torizon / Toradex BSP	HAVE	kit + SoM docs
UG10164 / HAB path	HAVE	ccli-nxp-yocto-ug
libiec61850	HAVE (refs)	GPLv3 - commercial to ship
meta-ccli layer	MISSING	create at freeze
Carrier device tree	MISSING	after pin budget
Modbus master + PF2 DO logic	NOT STARTED	needs analyzer map

6.1.14 To do

Ordered for **prototype BOM completeness** (Wave 1 SoM/kit **DONE**):

1. Download **IMX8MPIEC** (manual NXP) -> **reference/IMX8MPIEC.pdf**
2. Request Toradex **PCN / longevity letter** for PN **0063**
3. **Author CCI_Pin_Budget.md** (SoM -> UART/SPI/I2C/GPIO/RGMII -> connectors)
4. **Author CCI_Carrier_Block_Diagram.md** (freeze C1/C2/C4/C5)
5. Pick MPNs and download: **RS-485 isolator, DI opto, DO relay, Ethernet switch/PHY, GNSS module**
6. **Energy analyzer Modbus map** (plant under test)
7. Power tree IC datasheets (12-24 V in)
8. Altium schematic -> **ActiveBOM** export into this folder
9. (Optional) Compulab CL-SOM eval PDFs if second-source SoM needed

6.1.15 RAG / folder convention

```

knowledge-base/08-engineering/
  CCI_Prototype_BOM.md          -> ccli-prototype-bom
  CCI_Prototype_BOM.json
  CCI_Pin_Budget.md             -> ccli-pin-budget          (MISSING)
  CCI_Carrier_Block_Diagram.md  -> ccli-carrier-block        (MISSING)
  Toradex_Verdin_*.pdf          -> ccli-toradex-*            (HAVE)
  reference/
    IMX8MPCEC.pdf               -> ccli-nxp-imx8m-plus-cec    (HAVE)
    IMX8MPIEC.pdf               -> ccli-nxp-imx8m-plus-iec    (MISSING)
  vendor-validate/
    -> Wave 1 alternate/study PDFs (HAVE)
  <MPN>-datasheet.pdf           -> ccli-bom-<mpn-slug>       (carrier TBD)

```

Telemetry mirror: documents/projects/ccli/engineering/ (+ reference/, vendor-validate/).

6.1.16 Keywords

prototype bom, pin budget, carrier, relay, opto, RS-485, Eth_A, Eth_B, GNSS, LTE, Verdin, 99991105, 0063, Architecture Freeze, IMX8MPIEC, ccli-prototype-bom, Wave 1

III Regulations & protocols

Chapter 7

Module regulations classification

7.1 CCI Module - Hardware/Protocol Inventory & Regulations Classification

7.2 Modulo CCI - Inventario HW/protocolli e classificazione normativa

Doc ID: CCLI-REGS-CLASS-001 **Purpose / Scopo:** Classify every norm cited by the AiLux-class CCI module description into RAG categories, map features -> regulations -> `source_id`, and mark ingest status. **Audience / Destinatari:** Architecture freeze, certification planning, Telemetry RAG routing.

7.2.1 Italian <-> English glossary / Glossario IT <-> EN

Italiano (source)	English
CCI (Centralina / Controllore di Impianto)	Plant Controller / Central Plant Controller
DSO	Distribution System Operator
Operatore Abilitato (OA)	Qualified / Authorized Operator
Elementi di Impianto	Plant elements / DER field devices
Allegato O / Allegato T	Annex O / Annex T
Normative di riferimento	Reference standards / normative references
Cyber Security	Cybersecurity
Secur Boot (<i>datasheet spelling</i>)	Secure Boot
Anti-Tamper / antimanomissione	Anti-tamper / anti-tampering
Porte Seriali optoisolate	Opto-isolated serial ports
Consolle di Configurazione e Manutenzione Locale	Local configuration & maintenance console
Gestione Armamenti	Arming management (telecontrol arming)
Protocolli sicuri	Secure protocols
Ingressi Digitali / Uscite Digitali	Digital inputs (DI) / Digital outputs (DO)
da energizzare con alimentazione ESTERNA	to be energized from EXTERNAL field supply
Temperatura di esercizio	Operating temperature
Delibera ARERA	ARERA Decision / Resolution
conforme a	compliant with / conforms to

7.2.2 Module inventory (source description)

7.2.3 Inventario modulo (descrizione sorgente)

1.0 Full source text (IT) + English translation

Italiano (originale):

CPU iMX6D ARM Cortex-A9, 1GB DDR3, 4GByte eMMC, 800Mhz, CAAM, CSU, A-HAB SHA-256. Cyber Security: Secure Boot, Trusted Platform Module 2.0, FIPS 140-2 Level 3, RBAC, TRNG. Anti-Tamper hardware con switch collegato alla copertura e accelerometro X-Y-Z antimanomissione. Conforme a IEC-62443-4-1/2. Ethernet Switch a 4 porte 10/100 Mbps: una porta dedicata a DSO (ETH_A) e una porta dedicata a Operatore Abilitato (ETH_B); due porte per connessioni agli Elementi di Impianto; connettori RJ45 8/8. Due porte Seriali optoisolate RS-232/RS-485 su RJ45 8/8 standard ModBus. Una porta USB 2.0 per Consolle di Configurazione e Manutenzione Locale; due porte USB 2.0 per Elementi di Impianto. Protocolli seriali: IEC 61158-15 ModBus-RTU; IEC 60870-5-101 Balance/Unbalance; IEC 60870-5-103. Protocolli Ethernet: IEC 60870-5-104 Slave/Reverse/Redundancy (Gestione Armamenti Multicast/Unicast); IEC-61850 Server/Client/Goose; ModBus-TCP/IP Master/Slave; DNP3; S7-Comm; OPC-UA Client; C37.118 Client; IEC 62056-21 (DLMS/COSEM). Protocolli sicuri IEC 62351 parti 3-9 e 14. Servizi: http(s), sftp, ssh, snmp v2/v3 trap, ntpd, client/server PTP (IEEE-1588 power profile), redundancy cover system, firewall, RBAC, tasks ladder IEC 61131-3, web interface diagnostica/logs/SCADA. N°5 Ingressi Digitali optoisolati 10-120 Vdc (alimentazione esterna di campo); N°3 Uscite Digitali a relè 250 Vac 3A; GPS/GLONASS; modem 2G/3G/4G/LTE Cat.1; display OLED 1"; MicroSD fino a 32 GB; alimentazione 12-24 Vdc $\pm 20\%$ 5 W; temperatura di esercizio -5 °C... +45 °C. Normative di riferimento: CEI 0-16 allegati O-T; CEI EN 61557-12:2018; ARERA 540/2021; FIPS 140-2; IEC 62351-3/-4/-5/-6.

English:

CPU: i.MX6D ARM Cortex-A9, 1 GB DDR3, 4 GB eMMC, 800 MHz, CAAM, CSU, A-HAB SHA-256. Cybersecurity: Secure Boot, TPM 2.0, FIPS 140-2 Level 3, RBAC, TRNG. Hardware anti-tamper with cover-linked switch and X-Y-Z anti-tampering accelerometer. Compliant with IEC 62443-4-1/2. 4-port 10/100 Mbps Ethernet switch: one port dedicated to the DSO (ETH_A), one port dedicated to the Qualified/Authorized Operator (ETH_B); two ports for connections to plant elements (DER field devices); RJ45 8/8 connectors. Two opto-isolated serial ports RS-232/RS-485 on RJ45 8/8 Modbus pinout. One USB 2.0 port for local configuration & maintenance console; two USB 2.0 ports for plant elements. Serial protocols: IEC 61158-15 Modbus-RTU; IEC 60870-5-101 balanced/unbalanced; IEC 60870-5-103. Ethernet protocols: IEC 60870-5-104 Slave/Reverse/Redundancy (arming management via multicast and unicast); IEC 61850 Server/Client/GOOSE; Modbus-TCP/IP Master/Slave; DNP3; S7-Comm; OPC-UA Client; C37.118 Client; IEC 62056-21 (DLMS/COSEM). Secure protocols: IEC 62351 parts 3-9 and 14. Services: http(s), sftp, ssh, SNMP v2/v3 trap, ntpd, PTP client/server (IEEE 1588 power profile), redundancy cover system, firewall, RBAC, IEC 61131-3 ladder tasks, web UI for diagnostics/logs/SCADA. 5 opto-isolated digital inputs 10-120 Vdc (external field supply); 3 relay digital outputs 250 Vac / 3 A; GPS/GLONASS; 2G/3G/4G/LTE Cat.1 modem; 1" OLED; MicroSD up to 32 GB; supply 12-24 Vdc $\pm 20\%$, 5 W; operating temperature -5 °C... +45 °C. Reference standards: CEI 0-16 Annexes O-T; CEI EN 61557-12:2018; ARERA Decision 540/2021; FIPS 140-2; IEC 62351-3/-4/-5/-6.

1.1 Platform / CPU

Item (EN)	Voce (IT)	Spec
CPU	CPU	i.MX6D ARM Cortex-A9, 800 MHz
RAM / storage	RAM / memoria	1 GB DDR3, 4 GB eMMC
Crypto silicon	Silicio crittografico	CAAM, CSU, A-HAB SHA-256
Secure boot	Secure Boot	Secure Boot + HAB-class chain
TPM	TPM	TPM 2.0, FIPS 140-2 Level 3
Access control	Controllo accessi	RBAC
Entropy	Entropia	TRNG
IACS process	Processo IACS	Conforms to / conforme a IEC 62443-4-1 / 4-2
Anti-tamper	Anti-manomissione	Cover switch + X-Y-Z accelerometer

1.2 Networking & I/O

Item (EN)	Voce (IT)	Spec
Ethernet switch	Switch Ethernet	4x 10/100 Mbps, RJ45 8/8
ETH_A	ETH_A	Dedicated DSO / dedicata al DSO
ETH_B	ETH_B	Dedicated Qualified Operator (Operator Abilitato)
ETH plant	ETH impianto	2x plant elements / Elementi di Impianto
Serial	Porte seriali	2x optoisolated RS-232 / RS-485 on RJ45 8/8 Modbus pinout
USB	USB	1x USB 2.0 local config/maintenance console; 2x USB 2.0 plant elements
DI	Ingressi digitali	5x optoisolated 10-120 Vdc (external field supply)
DO	Uscite digitali	3x relay 250 Vac / 3 A
GNSS	GNSS	GPS/GLONASS, SMA-F, active antenna 5 V out, 5 m RG174
Cellular	Modem cellulare	Integrated 2G/3G/4G/LTE Cat.1, SMA-F
HMI	Display	OLED 1" local diagnostics
Storage	Storage	MicroSD <= 32 GB (logs/storage)
Power	Alimentazione	12-24 Vdc $\pm 20\%$, 5 W
Ambient	Temperatura di esercizio	-5 °C ... +45 °C

1.3 Serial protocols / Protocolli seriali (RS-232 / RS-485)

- IEC 61158-15 Modbus-RTU
- IEC 60870-5-101 Balanced / Unbalanced (*bilanciato / sbilanciato*)
- IEC 60870-5-103

1.4 Ethernet protocols / Protocolli Ethernet

- IEC 60870-5-104 Slave / Reverse / Redundancy - arming management / *Gestione Armamenti* (multicast & unicast)
- IEC 61850 Server / Client / GOOSE
- IEC 61158-15 Modbus-TCP/IP Master and/or Slave
- DNP3
- S7-Comm
- OPC-UA Client
- IEEE C37.118 Client
- IEC 62056-21 (DLMS/COSEM)

1.5 Secure protocol stack / Protocolli sicuri (IEC 62351)

Part	Scope EN (per module claim)	Ambito IT
IEC 62351-3	TLS for IEC 60870-5-101/104 and DNP3	TLS per 101/104 e DNP3
IEC 62351-4	Security for IEC 61850	Sicurezza per IEC 61850
IEC 62351-5	Security extensions for IEC 60870-5-101/104 and DNP3	Estensioni di sicurezza 101/104 e DNP3
IEC 62351-6	Security for IEC 61850 (GOOSE/SV related)	Sicurezza IEC 61850 (GOOSE/SV)
IEC 62351-7	SNMP network/system management security	Sicurezza SNMP
IEC 62351-8 / -9	Role-based access / key management for IEC 60870-5-101/104, IEC 61850, DNP3	RBAC / gestione chiavi
IEC 62351-14	Syslog	Syslog

1.6 Services / Servizi

http(s), sftp, ssh, SNMP v2/v3 trap, ntpd, PTP client/server (IEEE 1588 power profile), redundancy cover system, firewall, RBAC, IEC 61131-3 ladder tasks (*tasks ladder*), web UI (diagnostics / *diagnostica*, logs, SCADA).

1.7 Reference standards / Normative di riferimento (explicit on datasheet)

EN	IT
CEI 0-16 Annexes O-T	CEI 0-16 allegati O-T
CEI EN 61557-12:2018	CEI EN 61557-12:2018
ARERA Decision 540/2021	Delibera ARERA 540/2021
FIPS 140-2	FIPS 140-2
IEC 62351-3 / -4 / -5 / -6	IEC 62351-3 / -4 / -5 / -6

(IEC 62443-4-1/2 is claimed under Cybersecurity / Cyber Security, not only in the normative bullet list.)

7.2.4 Regulation taxonomy (RAG classes)

7.2.5 Tassonomia normativa (classi RAG)

Use these **class codes** when routing queries and when ingesting PDFs. Usare questi **codici classe** per il routing delle query e per l'ingest dei PDF.

Class EN	Classe IT	Code	Meaning EN / Significato IT
Italian grid / CCI mandate	Mandato rete IT / CCI	GRID-IT	Connection rules for DER plant controller (CCI) / Regole di connessione per il Controllore di Impianto
Market / regulator	Mercato / regolatore	REGULATOR-IT	ARERA decisions binding Italian DSO/plant ops / Delibere ARERA vincolanti
Measurement / PQ	Misura / qualità	METROLOGY	Performance measuring / monitoring equipment / Apparecchiature di misura e monitoraggio
IACS cybersecurity process & component	Cybersecurity IACS	CYBER-IACS	IEC 62443 product/process
Crypto module validation	Validazione modulo crypto	CYBER-CRYPTO	FIPS / TPM claims / Dichiarazioni FIPS/TPM
Power-system protocol security	Sicurezza protocolli power	CYBER-PROTO	IEC 62351 family / Famiglia IEC 62351
EU product cyber law	Normativa cyber UE prodotto	CYBER-EU	CRA (and related) / Cyber Resilience Act
EU energy policy	Politica energetica UE	ENERGY-EU	RED II (contextual) / Direttiva rinnovabili
Field / SCADA protocols	Protocolli di campo / SCADA	PROTO-FIELD	60870 / 61850 / Modbus / DNP3 / ...
Time sync	Sincronizzazione oraria	TIME	NTP / PTP / GNSS
PLC / application	PLC / applicazione	APP-PLC	IEC 61131-3
ATEX / hazardous area	ATEX / area pericolosa	ATEX	Existing Telemetry corpus (not CCI datasheet primary)
Product EMC / radio	EMC / radio prodotto	EMC-RADIO	Cellular modem / emissions (if certified later)

7.2.6 Classification table - every cited norm

7.2.7 Tabella di classificazione - ogni norma citata

Norm / Norma	Class	Applies to EN / IT	Priority	RAG status	source_id (target or actual)
CEI 0-16 Annex O / Allegato O	GRID-IT	CCI functional / plant controller role	P0	INGESTED	cei-0-16-allegato-o
CEI 0-16 Annex T / Allegato T	GRID-IT	IEC 61850 exchange with DSO / scambio IEC 61850 con DSO	P0	INGESTED	cei-0-16-allegato-t
AiLux CCI manual (impl. baseline)	GRID-IT	Full module behaviour reference / riferimento comportamento modulo	P0	INGESTED	cei-0-16-cci-manual-ailux
ARERA Decision / Delibera 540/2021	REGULATOR-IT	Italian market obligations tied to CEI 0-16 CCI	P0	MISSING	arera-540-2021
CEI EN 61557-12:2018	METROLOGY	Performance measuring / monitoring claims	P1	MISSING	cei-en-61557-12-2018
IEC 62443-4-1	CYBER-IACS	Secure development lifecycle	P0	MISSING	iec-62443-4-1
IEC 62443-4-2	CYBER-IACS	Technical security requirements for components	P0	MISSING	iec-62443-4-2
FIPS 140-2 (L3 TPM claim)	CYBER-CRYPTO	TPM 2.0 / crypto boundary	P1	MISSING	fips-140-2
IEC 62351-3	CYBER-PROTO	TLS for 101/104, DNP3	P0	MISSING	iec-62351-3
IEC 62351-4	CYBER-PROTO	IEC 61850 security	P0	MISSING	iec-62351-4
IEC 62351-5	CYBER-PROTO	101/104 & DNP3 security	P0	MISSING	iec-62351-5
IEC 62351-6	CYBER-PROTO	IEC 61850 GOOSE/SV security	P0	MISSING	iec-62351-6
IEC 62351-7	CYBER-PROTO	SNMP security	P1	MISSING	iec-62351-7
IEC 62351-8	CYBER-PROTO	RBAC for power protocols	P1	MISSING	iec-62351-8
IEC 62351-9	CYBER-PROTO	Key management	P1	MISSING	iec-62351-9
IEC 62351-14	CYBER-PROTO	Syslog	P2	MISSING	iec-62351-14
Regulation (EU) 2024/2847 CRA	CYBER-EU	Products with digital elements (clone path)	P1	INGESTED	eu-2024-2847-cra
Directive (EU) 2018/2001 RED II	ENERGY-EU	Renewables policy context	P2	INGESTED	eu-2018-2001-red
IEC 60870-5-101	PROTO-FIELD	Serial telecontrol	P1	MISSING	iec-60870-5-101
IEC 60870-5-103	PROTO-FIELD	Protection equipment serial	P2	MISSING	iec-60870-5-103
IEC 60870-5-104	PROTO-FIELD	Ethernet telecontrol + arming	P1	MISSING	iec-60870-5-104
IEC 61850 (core + GOOSE)	PROTO-FIELD	Server/Client/GOOSE	P0	PARTIAL via Annex T	iec-61850 (target); today use cei-0-16-allegato-t
IEC 61158-15 / Modbus	PROTO-FIELD	RTU + TCP	P1	MISSING	iec-61158-15-modbus
IEEE 1815 / DNP3	PROTO-FIELD	DNP3 (+ 62351-3/5)	P2	MISSING	ieee-1815-dnp3
OPC UA (IEC 62541)	PROTO-FIELD	OPC-UA Client	P2	MISSING	opc-ua-iec-62541
IEEE C37.118	PROTO-FIELD	Synchrophasor client	P2	MISSING	ieee-c37-118
IEC 62056-21 DLMS/COSEM	PROTO-FIELD	Metering serial/optical profile	P2	MISSING	iec-62056-21

7.2.8 Feature -> regulation map (for RAG queries)

7.2.9 Mappa funzione -> normativa (per query RAG)

CCI feature cluster (EN / IT)	Primary classes	Preferred source_ids (available now)	Preferred when in- gested
ETH_A DSO / ETH_B OA (Opera- tore Abilitato) / plant ports (Elementi di Impianto)	GRID-IT	cei-0-16-allegato-o, cei-0-16-allegato-t, cei-0-16-cci-manual-ailux	arera-540-2021
IEC 61850 / GOOSE	GRID-IT, PROTO-FIELD, CYBER-PROTO	cei-0-16-allegato-t	iec-61850, iec-62351-4, iec-62351-6
IEC 104 arming multi- cast/unicast	PROTO-FIELD, CYBER-PROTO	cei-0-16-cci-manual-ailux	iec-60870-5-104, iec-62351-3, iec-62351-5
Modbus RTU/TCP	PROTO-FIELD	cei-0-16-cci-manual-ailux	iec-61158-15-modbus
Secure Boot / HAB / CAAM / TPM / FIPS	CYBER-CRYPTO, CYBER-IACS	this class doc + roadmap	fips-140-2, iec-62443-4-1, iec-62443-4-2
Anti-tamper + RBAC + firewall	CYBER-IACS, CYBER-PROTO	this class doc	iec-62443-4-2, iec-62351-8
SNMP / Syslog secure	CYBER-PROTO	this class doc	iec-62351-7, iec-62351-14
DI/DO / GPS / LTE / OLED / microSD	product HW (manual)	cei-0-16-cci-manual-ailux	
Metrology / monitor- ing accuracy	METROLOGY	-	cei-en-61557-12-2018
Clone product CE cy- ber	CYBER-EU	eu-2024-2847-cra	+ 62443 / 62351 cor- pus
Renewables policy con- text	ENERGY-EU	eu-2018-2001-red	-

7.2.10 RAG routing packs (copy into chat / query filters)

7.2.11 Pacchetti di routing RAG (da usare in chat / filtri query)

Pack A - Italian CCI mandate / Mandato CCI italiano (available now)

cei-0-16-allegato-o, cei-0-16-allegato-t, cei-0-16-cci-manual-ailux, ccli-cci-module-regs-class

Pack B - Cyber for CCI clone (partial now)

ccli-cci-module-regs-class, eu-2024-2847-cra, cei-0-16-cci-manual-ailux

Add when PDFs arrive: iec-62443-4-1, iec-62443-4-2, iec-62351-3, iec-62351-4, iec-62351-5, iec-62351-6, fips-140-2.

Pack C - Protocols (partial now)

ccli-cci-module-regs-class, cei-0-16-allegato-t, cei-0-16-cci-manual-ailux

Pack D - Market + metrology / Mercato + metrologia (blocked until download)

arera-540-2021, cei-en-61557-12-2018

7.2.12 To do (regulation PDFs) / Backlog acquisizione (P0 first)

1. ARERA Decision / Delibera 540/2021 (and any later amending decisions you use)
2. IEC 62443-4-1 and IEC 62443-4-2
3. IEC 62351-3, -4, -5, -6 (datasheet “Normative di riferimento” set)
4. CEI EN 61557-12:2018
5. FIPS 140-2 (or FIPS 140-3 transition note if TPM cert is 140-3)
6. Protocol base standards as needed for implementation (60870-5-104, 61850 core, Modbus)

Folder convention under Telemetry:

```
documents/regulations/
  cei-0-16/...           # already present
  eu/...                 # RED + CRA present
  arera/arera-540-2021/
  cei-en/cei-en-61557-12-2018/
  iec-62443/iec-62443-4-1/
  iec-62443/iec-62443-4-2/
  iec-62351/iec-62351-3/ ... -6/ (and optional -7/-8/-9/-14)
  fips/fips-140-2/
  protocols/iec-60870-5-104/ ...
```

Target `source_id` = folder leaf name (see Sec.3).

7.2.13 Notes / caveats / Note

- **IEC 61158-15** is the fieldbus family reference often used commercially for Modbus; confirm exact Modbus.org / IEC edition when purchasing standards.
- **S7-Comm** is proprietary - no public IEC PDF expected; keep as protocol capability only.
- **FIPS 140-2 Level 3** is a *crypto module* claim (typically TPM); do not treat as whole-device FIPS certification unless a certificate number is provided.
- **IEC 62443-4-1/2 “conforme” / “compliant with”** on a competitor datasheet is a claim - clone path needs evidence (SDL artifacts + component requirements) before CE/CRA arguments.
- This classification document is **not** a substitute for licensed standard PDFs; it routes RAG until those PDFs are ingested. / Questo documento **non** sostituisce i PDF delle norme: serve al routing RAG finché non sono ingeriti.

IV Engineering infrastructure

Chapter 8

Industrial RAG architecture

8.1 CCI Industrial RAG Architecture (three indexes)

Doc ID: CCLI-RAG-ARCH-001 **Revision:** 1.0 **Date:** 2026-07-25 **Target score:** ~9.8/10 industrial engineering RAG (from ~8.5/10 vector-first baseline)

8.1.1 Three indexes

Index	Role	Where
Vector	Semantic similarity over chunk embeddings	Telemetry SQLite <code>chunks.embedding_json</code> (+ optional Qdrant)
Keyword	Exact token / acronym retrieval (HAB, TPM, IEC62351, 99991105...)	Token overlap + exact-term boost; optional Elasticsearch BM25
Knowledge graph	Typed reasoning paths	Seed JSON + SQLite <code>graph_edges</code>

Hybrid fusion (query time):

```
score = 0.50 x vector + 0.50 x keyword
      + chunk_type_boost + context_boost + entity_overlap_boost
      + exact_industrial_term_boost (capped)
```

Then attach **seed graph paths** and **missing_sources** (from regulatory map) into query traces / citation `graph_paths`.

8.1.2 Artifacts

Artifact	Path
Seed knowledge graph	08-engineering/cci-knowledge-graph.json
Entity lexicon (CCI)	Telemetry rag-python/car_manual_pipeline/dictionaries/ccli_entities
Regulatory routing	07-protocols/CCI_Module_Regulations_Classification.md (+ .json)
Eval set	08-engineering/cci-hybrid-rag-eval.json
Entity extract	entity_extraction.py (loads car + CCI dictionaries)
Graph expand	graph_rag.py (seed_graph_paths_for_query, missing_sources_for_query)
Hybrid retrieve	retrieval.py (POST /v1/query)

Env override for graph path: CCI_KNOWLEDGE_GRAPH_PATH.

8.1.3 Edge types (seed graph)

ENABLES · IMPLEMENTS · REQUIRES · MAPS_TO · COVERED_BY · **PARTIAL** · **MISSING** · CERT_CLAIM
· RELATED Example spine:

```
i.MX 8 -> TrustZone -> Secure Boot -> HAB
Eth_A -> IEC 61850 -> IEC 62351
Verdin 99991105 -[PARTIAL]-> C1 Ethernet
Verdin 99991105 -[MISSING]-> C2 RS-485 / C4 DI/DO / C5 GNSS
```

8.1.4 Query contract (already on /v1/query)

Request: tenant_id, bot_id, query, source_ids[], top_k, grounding_prompt. Response traces now include:

- hybrid_mode
- query_entities
- seed_graph_paths
- missing_sources (**MISSING**/**PARTIAL** norms from seed rag_status_refs)
- Citations carry graph_paths + chunk_metadata.entity_names (after re-ingest)

8.1.5 Apply after code deploy

1. Restart RAG container / process so new dictionaries and scoring load.
2. Re-ingest CCLI sources so chunk metadata gets CCI entities:

```
cd "C:\Yahia\projects\Telemetry System"
.\scripts\ingest-ccli.ps1
```

1. Smoke-test:

```
$body = @{
    tenant_id = "telemetry"; bot_id = "default"
    query = "What CCI controls does Toradex Verdin 99991105 not cover regarding HAB and RS-485?"
    source_ids = @("ccli-vendor-selection","ccli-toradex-verdin-devboard","ccli-cci-module-regs-
    top_k = 8; answer_language = "en"; clarify = $false
} | ConvertTo-Json
Invoke-RestMethod http://localhost:8001/v1/query -Method POST -Body $body -ContentType "applic
    Select-Object -ExpandProperty traces | ConvertTo-Json -Depth 5
```

Expect: query_entities containing hab / verdin / rs485, seed_graph_paths with **PARTIAL**/**MISSING** edges, missing_sources empty or limited unless norms are asked.

8.1.6 Still out of MVP (next)

- Full Okapi BM25 / SQLite FTS5 inverted index (today: overlap + exact boost + optional ES)
 - Semantic re-chunking pass (heading/table-aware) beyond current classifiers
 - LLM entity extraction second pass
 - Graph DB (Neo4j) - seed JSON is enough until scale
-

8.1.7 Keywords

hybrid rag, bm25, knowledge graph, entity extraction, HAB, TrustZone, IEC 62351, CEI 0-16, ccli-knowledge-graph, industrial rag

Chapter 9

Master download index

9.1 CCI Knowledge Base - Master Download Index

Document ID: CCLI-KB-DL-001 **Revision:** 1.0 **Date:** 2026-07-25 **Root on disk:** c:\Yahia\projects\CCLI\knowledge-base\dl-001
Telemetry mirror: C:\Yahia\projects\Telemetry System\documents\projects\ccli\ Use this file as the single map of **local path** <-> **source URL** <-> **RAG source_id**.

9.1.1 Yocto / BitBake (01-yocto/, 02-bitbake/)

Local path	Download / source URL	RAG source_id	Status
01-yocto/yocto-5.0.13-theyoctoproject.pdf	https://yoctoproject.org/5.0.13/_static/the-yoctoproject.pdf	ccli-yocto-overview	ON DISK + INGESTED
01-yocto/yocto-5.0-scarthgap-theyoctoproject.pdf	https://yoctoproject.org/scarthgap/_static/the-yoctoproject.pdf - (tip mirror)		ON DISK
02-bitbake/bitbake-user-manual-scarthgap-era-extracting-bitbake/	https://yoctoproject.org/scarthgap-era-extracting-bitbake/	ccli-bitbake-manual	ON DISK + INGESTED
01-yocto/NXP_UG10164_iMX_Yocto_Users_Guide.pdf	Reference: https://docs.nxp.com/bundle/UG10164/PDF:https://www.nxp.com/docs/en/user-guide/UG10164.pdf	ccli-nxp-yocto-ug	ON DISK + INGESTED - official PDF at 01-yocto/NXP_UG10164_iMX_Yocto_P
01-yocto/CCI_GitHub_Yocto_References.md	CCCI upstream docs harvest (see Sec.2)	ccli-github-yocto-ref	ON DISK + INGESTED
01-yocto/CCI_GitHub_Paths_and_Layers_Files.md	CCCI upstream SHA catalog	ccli-github-layer-pin	ON DISK + INGESTED
01-yocto/github-refs/*.md	Individual upstream READMEs	(included in combined)	ON DISK
01-yocto/DOWNLOADS.md	Yocto-only subset of this index	-	ON DISK

Pin note: CCI upstream docs pin **Yocto 5.0.13 Scarthgap**. NXP BSP tip is **LF6.18.20 / Wrynose** via [imx-manifest](<https://github.com/nxp-imx/imx-manifest>) - reconcile with Toradex/Variscite BSP at SoM freeze.

Manual PDF drop (NXP UG10164)

After downloading from NXP (browser login may be required):

```
Copy-Item "$env:USERPROFILE\Downloads\UG10164.pdf" 'c:\Yahia\projects\CCLI\knowledge-base\01-yocto\NXP_UG10164_iMX_Yocto_Project_Users_Guide.pdf'
```

9.1.2 GitHub references (01-yocto/github-refs/)

Local path	Upstream URL	
github-refs/nxp-imx-manifest-wrynose.md	https://github.com/nxp-imx/imx-manifest (branch <code>imx-linux-wrynose</code>)	
github-refs/nxp-imx-manifest-scarthgap.md	https://github.com/nxp-imx/imx-manifest (branch <code>imx-linux-scarthgap</code>)	
github-refs/meta-freescale-scarthgap.md	https://github.com/Freescale/meta-freescale (branch <code>scarthgap</code>)	
github-refs/libiec61850.md	https://github.com/mz-automation/libiec61850 (GPLv3 + commercial)	
github-refs/lib60870.md	https://github.com/mz-automation/lib60870	
github-refs/meta-secure-core.md	https://github.com/jiazhang0/meta-secure-core	
github-refs/toradex-signed-boot.md	https://github.com/toradex/build-signed-imx8m-bootloader	
01-yocto/CCI_GitHub_Paths_and_Layer_Pins.md	undored SHA/layer catalog)	ccli-github-layer-pins
github-refs/meta-toradex-security.md	https://github.com/toradex/meta-toradex-security	
github-refs/meta-imx-scarthgap.md	https://github.com/nxp-imx/meta-imx	

Also track (catalogued in `CCI_GitHub_Paths_and_Layer_Pins.md`):

Repo	URL	When
meta-imx	https://github.com/nxp-imx/meta-imx	After SoM freeze
meta-variscite-imx	https://github.com/varigit/meta-variscite-imx	If Variscite chosen
variscite-bsp-platform	https://github.com/varigit/variscite-bsp-platform	If Variscite chosen
meta-bsp-imx8mm	https://github.com/compulab-yokneam/meta-bsp-imx8mm	If Compulab chosen
meta-security	https://git.yoctoproject.org/meta-security	Hardening

9.1.3 Engineering / Toradex kit (08-engineering/)

Local path	Download URL	RAG source_id	Status
08-engineering/CCI_Vendor_Selection.md	action.md	ccli-vendor-selection	INGESTED
08-engineering/CCI_Validation_and_Mocking_Strategy.md	action.md	ccli-validation-strategy	ON DISK + INGESTED
08-engineering/CCI_Mocking_Bench_BOM.md	action.md	ccli-mocking-bench-bom	ON DISK + INGESTED
08-engineering/CCI_RAG_Knowledge_Base.md	action.md	ccli-rag-knowledge-base	ON DISK + INGESTED
08-engineering/CCI_Mocking_Bench_BOM.json	action.md	-	ON DISK
08-engineering/Toradex_Verdin_Development_Board_V1m1_Datasheet.pdf	https://www.toradex.com/en/development-board-datasheet.pdf	ccli-toradex-verdin-development-board	ON DISK
08-engineering/Toradex_Verdin_Carrier_Board_Design_Guide.pdf	https://www.toradex.com/en/carrier-board-design-guide.pdf	ccli-toradex-carrier-board	ON DISK
08-engineering/Toradex_Verdin_Family_Specification.pdf	https://www.toradex.com/en/verdin-family-specification.pdf	ccli-toradex-verdin-family	ON DISK
08-engineering/Toradex_Verdin_IMX8M_Plus_1T2_Datasheet.pdf	https://www.toradex.com/en/imx8m-plus-1t2-datasheet.pdf	ccli-toradex-verdin-imx8m-plus-1t2	ON DISK + INGESTED

Product page (kit PN 99991105): <https://www.toradex.com/products/carrier-board/verdin-development-board-kit> **Developer center:** <https://developer.toradex.com/hardware/verdin-som-family/carrier-boards/verdin-development-board>

9.1.4 3b. NXP SoC datasheets (08-engineering/reference/)

Mirrored under Telemetry projects/ccli/reference/.

Local path	Download / source URL	RAG source_id	Status
08-engineering/reference/IMX8MPCEC.pdf	IMX8MPCEC (i.MX 8M Plus consumer / industrial CEC)	ccli-nxp-imx8m-plus-cec	ON DISK + IN-GESTED
08-engineering/reference/IMX8MPLUS.pdf	IMX8MPLUS https://www.nxp.com/products/i.MX8MPLUS (industrial IEC)	ccli-nxp-imx8m-plus-iec	MISSING
08-engineering/reference/IMX8MDQLQCEC.pdf	IMX8MDQLQCEC (i.MX 8M Dual/QuadLite)	ccli-nxp-imx8m-dqlq-cec	ON DISK + IN-GESTED

Precision extract (authored): CCI_NXP_IMX8MDQLQCEC_Precision_Extract.md -> ccli-nxp-imx8m-dqlq-cec-pr

9.1.5 Protocols / EU regulations (07-protocols/)

Local path	Source	RAG source_id	Status
07-protocols/CCI_Module_Regulations_Classification.md	CCI_Module_Regulations_Classification	ccli-cci-module-regs	IN-GESTED
07-protocols/CCI_Module_Regulations_Classification.json	CCI_Module_Regulations_Classification		ON DISK
07-protocols/CELEX_320182001-Directive-2018/2001	CELEX_320182001-Directive-2018/2001	eu-2018-2001-red	ON DISK (also Telemetry regulations/eu/"!)
07-protocols/OJ_L_20242847-Regulation-2024/2847-cra	OJ_L_20242847-Regulation-2024/2847-cra	eu-2024-2847-cra	ON DISK (also Telemetry)

Telemetry regulation paths:

C:\Yahia\projects\Telemetry System\documents\regulations\eu\directive-2018-2001-red\CELEX_320182001-Directive-2018/2001
C:\Yahia\projects\Telemetry System\documents\regulations\eu\regulation-2024-2847-cra\OJ_L_20242847-Regulation-2024/2847-cra
C:\Yahia\projects\Telemetry System\documents\regulations\cei-0-16\allegato-o\
C:\Yahia\projects\Telemetry System\documents\regulations\cei-0-16\allegato-t\
C:\Yahia\projects\Telemetry System\documents\regulations\cei-0-16\manuale-cci\

9.1.6 Repo-root project docs (mirrored to Telemetry projects/ccli/)

Local path (CCLI repo)	RAG source_id
CCI_Knowledge_Base_Brief.md (= knowledge-base/README.md)	ccli-knowledge-base-brief
CCI_Project_Roadmap.md / .tex / .pdf	ccli-project-roadmap
CCI_SoM_Comparison.tex / .pdf	ccli-som-comparison
CCI_Clone_RE_Option.tex / .pdf	ccli-clone-re-option
CCI_Yocto_BSP_Notes.md	ccli-yocto-bsp-notes
CCI_NXP_IMX8MDQLQCEC_Precision_Extract.md	ccli-nxp-imx8m-dqlq-cec-precise
08-engineering/reference/IMX8MDQLQCEC-1280316.pdf	ccli-nxp-imx8m-dqlq-cec
08-engineering/reference/IMX8MPCEC.pdf	ccli-nxp-imx8m-plus-cec

9.1.7 To do (downloads)

#	Item	Suggested URL / source	Target local path	Target source_id
1	NXP UG10164 official PDF	https://www.nxp.com/docs/en/user-guide/UG10164.pdf	01-yocto/NXP_UG10164_iMX_Yocto_Project_Users_Guide.pdf	DONE
2	Verdin iMX8M Plus IT SoM datasheet	https://docs.toradex.com/116795-verdin_imx8mp_datasheet.pdf	08-engineering/Toradex_Verdin_iMX8M_Plus_IT_SoM_Datasheet.pdf	DONE
3	ARERA 540/2021	ARERA website	07-protocols/ARERA_540-2021.pdf	DONE
4	CEI EN 61557-12:2018	CEI / licensed	07-protocols/CEI_EN_61557-12:2018.pdf	DONE
5	IEC 62443-4-1 / 4-2	IEC / licensed	07-protocols/iec-62443-4-1 / -4-2	DONE
6	IEC 62351-3-6	IEC / licensed	07-protocols/iec-62351-3-6	DONE
7	Variscite / Compulab remaining PDFs	Symphony V2, CL-SOM EVK, IMX8MPIEC	08-engineering/ ccli-variscite-* / ccli-compulab-*	

Validated vendor PDFs (2026-07-25)

Local path	Status	Notes
08-engineering/Toradex_Verdin_iMX8M_Plus_IT_SoM_Datasheet.pdf	INGESTED	ccli-toradex-verdin-plus-som
08-engineering/reference/vendor/Variscite/DART-MX8M-Plus-Datasheet.pdf	INGESTED	ccli-variscite-dart-mx8m-plus
08-engineering/reference/vendor/Variscite/DT8M-CustomBoard-Datasheet.pdf	INGESTED	ccli-variscite-dt8m-customboard
08-engineering/reference/vendor/Compulab/IOT-GATE-iMX8-Reference-Guide.pdf	INGESTED	ccli-compulab-iot-gate-ref (study)

9.1.8 Re-ingest helper (Telemetry)

```
cd "C:\Yahia\projects\Telemetry System"
.\scripts\ingest-document.ps1 -SourceId "ccli-nxp-yocto-ug" '
-PdfPath ".\documents\projects\ccli\yocto\NXP_UG10164_iMX_Yocto_Project_Users_Guide.pdf" -KeepInPlace
.\scripts\ingest-document.ps1 -SourceId "ccli-github-yocto-refs" '
-PdfPath ".\documents\projects\ccli\yocto\CCI_GitHub_Yocto_References.md" -KeepInPlace
.\scripts\ingest-document.ps1 -SourceId "ccli-toradex-verdin-devboard" '
-PdfPath ".\documents\projects\ccli\engineering\Toradex_Verdin_Development_Board_V1.1_Datash
```

Full CCLI set: `.\scripts\ingest-ccli.ps1`

V Validation & test

Chapter 10

PF2 validation & mocking strategy

10.1 CCI Validation & Mocking Strategy - PF2 Bench Ladder

Document ID: CCLI-VAL-001 **Revision:** 1.0 **Date:** 2026-07-25 **RAG source_id:** ccli-validation-strategy
Purpose: Define a progressive, evidence-based bench strategy for validating Modbus metering, PF2 logic, IEC 61850/104, digital I/O, logging, and safe fault handling before plant deployment.
Companions: CCI_Mocking_Bench_BOM.md | CCI_Prototype_BOM.md | CCI_Vendor_Selection.md | CCI_Project_Roadmap.md

This ladder validates engineering behavior; it does **not** replace accredited CEI 0-16 demonstrations, IEC 61850 conformance testing, EMC/safety testing, or IEC 62443 product/process assessment.

10.1.1 Recommended progression

Level	Bench	Main value	Phase 1 gate
L1 - Software mock	PC-hosted Modbus/IEC protocol simulators	Fast, deterministic application tests	Required first
L2 - Interface HIL	Controller + physical Ethernet/RS-485 + emulator PC	Drivers, timing, framing, cable faults	Recommended Phase 1
L3 - Real analyzer	Controller + selected energy analyzer	Vendor register map and device behavior	Required before field pilot
L4 - Power-system HIL	Typhoon HIL / OPAL-RT / RTDS-class real-time simulator	Closed-loop grid/inverter/protection behavior	Later system-validation campaign

Do not jump directly to L3 or L4. Keep the same versioned test vectors and expected outcomes across all levels so failures can be isolated to software, interface, device integration, or closed-loop behavior.

10.1.2 Level 1 - software-only mocking

Run a Modbus server on an engineering PC. The controller polls it using the same application code, register map abstraction, scaling, timeout policy, and PF2 state machine intended for the product.

Engineering PC

```

+-- Modbus TCP/RTU simulator
+-- P = 850 kW
+-- Q = 100 kvar
+-- power factor = 0.98

```

Candidate tools: **pymodbus simulator**, **ModbusPal**, **QModMaster**, and **ModRssim2**. Tool selection is not frozen; record the exact version and configuration in each test report.

2.1 Minimum scenarios

Test ID	Stimulus	Expected controller behavior
L1-P-01	P = 500 kW	No curtailment
L1-P-02	P = 700 kW	No curtailment
L1-P-03	P = 950 kW	Curtailment request after configured debounce
L1-P-04	P = 1,000 kW	Curtailment remains active; no repeated command storm
L1-C-01	Meter timeout	Quality invalid, alarm/event, configured safe state
L1-C-02	Illegal function / exception response	Diagnostic event; no stale value treated as valid
L1-C-03	Bad CRC (RTU)	Frame rejected and counted; retry policy applied
L1-D-01	Out-of-range / NaN-equivalent register pattern	Input rejected or clamped per ICD; alarm raised
L1-R-01	Process restart	State recovery follows persistence and fail-safe policy

Important: bad-CRC and malformed-frame injection may require a purpose-built serial fault proxy or test script; many general Modbus simulators only emit valid frames.

2.2 PF2 reference sequence

Illustrative rule only: **if valid active power $P > 900$ kW, request curtailment.**

Time	P	Expected
00:00	500 kW	No action
00:30	700 kW	No action
01:00	950 kW	Curtail after debounce
01:30	1,000 kW	Curtail remains active

The production test must also define hysteresis, debounce, minimum on/off time, stale-data timeout, command acknowledgement, manual override, restart behavior, and release threshold. A single $P > 900$ comparison is not sufficient for a safe controller. Verify every transition through:

- PF2 state-machine state and reason code
- digital-output intent (mocked at L1)
- IEC 61850 / IEC 104 / Modbus command intent, where applicable
- alarm generation and acknowledgement
- timestamped event and measurement logs
- input quality, source, engineering units, and scaling

10.1.3 Level 2 - physical-interface HIL (recommended)

Use the real controller serial and Ethernet interfaces while the plant-side device remains emulated.

CCI controller

```
|
+-- RS-485 cable -- USB-to-RS-485 adapter -- Linux PC emulator
+-- Ethernet ----- engineering switch ----- protocol test client
```

This validates:

- Linux UART/serial driver and device-tree mapping
- RS-485 direction control, polarity, termination, and bias
- configured baud rate, parity, stop bits, slave address, retries, and timeout
- real frame timing and recovery after disconnect/reconnect
- Ethernet interface binding and Eth_A/Eth_B policy
- controller CPU/load behavior under protocol traffic

3.1 C2 limitation on lead eval hardware

The Toradex Verdin Development Board **99991105** provides **one non-isolated RS-485 DB9 path**. It is suitable for early L2 software/driver testing, but it does not prove the product requirement of **two opto-isolated RS-485 RJ45-class plant ports**. Repeat the electrical tests on the custom carrier.

3.2 L2 fault injection

- unplug/reconnect RS-485 while polling
- reverse A/B during a controlled test
- remove termination or introduce a long/noisy cable
- use wrong baud/parity/slave ID
- delay responses beyond timeout
- disconnect Eth_A, Eth_B, and plant Ethernet independently
- restart the emulator while the controller remains online

For each case verify bounded retries, no task deadlock, invalid data quality, event logging, alarm lifecycle, and a defined safe state.

10.1.4 Level 3 - real energy analyzer

Connect the selected production-candidate analyzer, for example a Schneider PM, Janitza UMG, Carlo Gavazzi, or Siemens PAC family device. The exact model and firmware must be frozen in the test record. A real analyzer validates:

- vendor register addresses, word order, signedness, scaling, and units
- update rate and communication latency
- device exception behavior and configuration persistence
- voltage, current, active/reactive/apparent power, frequency, power factor, and quality flags

No high-power plant is required for communications testing. Use the analyzer's supported test/simulation mode, low-voltage injection equipment, or a calibrated secondary-injection source as appropriate. **Do not assume arbitrary register values can be manually configured on every analyzer.** Confirm the chosen model's test mode and obtain its official **Modbus register map**; this remains a P1 missing document in the prototype BOM.

4.1 L3 acceptance gate

- official register map is on disk and ingested
 - every consumed register has address, type, scale, unit, validity rule, and polling rate
 - controller readings agree with analyzer display/reference within declared tolerance
 - loss/recovery and analyzer reboot tests pass
 - firmware/model-specific deviations are documented
-

10.1.5 Level 4 - real-time power-system HIL

Use a Typhoon HIL, OPAL-RT, RTDS, or equivalent real-time platform when closed-loop interaction with a modeled grid, inverter, transformer, protection relay, and communications network is justified. The controller may receive and emit:

- IEC 61850 MMS, reports, and GOOSE
- IEC 60870-5-104
- Modbus RTU/TCP
- DI/DO and analog/measurement signals through suitable I/O interfaces

Use L4 to test dynamic ramps, oscillation, command delays, protection events, communications congestion, failover, and interactions among PF2 logic, inverter controls, and DSO/operator commands. L4 is valuable but expensive. It is **not required to start Phase 1**, and it does not itself certify CEI 0-16 or IEC 61850 conformance.

10.1.6 Digital-input test box (C4)

Product target: **5 opto-isolated DI channels, 10-120 Vdc, external field supply**. Use a guarded, fused test fixture with selectable supported voltages (typically 24 V, 48 V, and 110 V where the final input design permits). Inject named states such as breaker closed/open, alarm active, meter fault, and external permissive. Verify:

- asserted/deasserted thresholds and hysteresis
- polarity protection and channel isolation
- debounce and contact chatter behavior
- Linux GPIO/application state mapping
- event timestamp and source identity
- simultaneous-channel transitions
- open-wire/power-loss behavior where detectable

Safety: 110 Vdc is hazardous. Use current limiting, fusing, guarded terminals, an emergency disconnect, and competent personnel. Start at 24 V until the carrier input circuit has passed design review.

10.1.7 Dry-contact output tests (C4)

Connect each relay output to a safe indicator lamp, PLC input, or isolated digital-input module. Do not switch mains loads during early bench testing.

PF2 active -> output policy permits actuation -> relay closes -> lamp/PLC input ON

Measure:

- command-to-contact response time

- contact bounce and application filtering
- power-up, reboot, watchdog, and power-loss state
- normally-open/normally-closed interpretation
- stuck relay / feedback mismatch handling if feedback is available
- manual override and command interlock behavior

The safe state must be specified by the system safety/control analysis; "relay open" is not automatically safe for every plant.

10.1.8 IEC 61850 and IEC 104 bench

```
IED test client / protocol emulator
      |
      Ethernet switch
      |
      CCI controller
```

Test MMS reads, control writes with the configured control model, buffered/unbuffered reports, GOOSE publish/subscribe, dataset/configuration revision handling, time quality, disconnect/reconnect, and malformed or unauthorized requests. Candidate tools include **Omicron IEDScout**, **Omicron StationScout**, and test clients built from **libiec61850**. libiec61850 is suitable for prototyping under GPLv3; a commercial license is required for a closed shipped product. For IEC 60870-5-104, test connection supervision, sequence counters, time-tagged events, command select/execute if used, spontaneous transmissions, and reconnect behavior.

10.1.9 Cross-interface failure matrix

Failure	Required checks
Meter disconnected / RS-485 cut	Invalid quality, alarm, event, bounded retry, safe PF2 behavior
Eth_A disconnected	DSO link alarm; plant functions remain bounded
Eth_B disconnected	Operator link alarm; no unauthorized fallback
GNSS lost	Time-quality downgrade; holdover policy; event
LTE lost	Backup-link alarm; primary control path unaffected
Process crash / watchdog reset	Controlled restart; outputs go to defined state
Stale measurement	Never treated as current; command inhibition or fallback per policy
Conflicting DSO/operator/local command	Deterministic priority and audit trail

The most important acceptance criterion is not "an alarm appeared"; it is that a communication fault cannot cause uncontrolled curtailment or silent use of stale data.

10.1.10 Practical Phase 1 PF2 bench

```
Engineering PC
+-- IEC 61850 / IEC 104 client and traffic capture
+-- Modbus simulator and test-vector runner
+-- USB-to-RS-485 adapter
```

```

      |
Ethernet switch
      |
CCI controller / Verdin kit
+-- DI test box (24 V first)
+-- DO test lamps / isolated PLC inputs

```

This bench can validate most application behavior without a solar plant, inverter, or grid connection:

- Modbus RTU/TCP acquisition
- PF2 state machine and command arbitration
- IEC 61850 and IEC 104 application behavior
- DI/DO mapping on suitable hardware
- logging, alarms, time quality, and fault handling
- basic cybersecurity controls and interface separation

It cannot prove final carrier isolation/EMC, analyzer accuracy, full power-system dynamics, or product conformance.

10.1.11 Test evidence required

Every test case should record:

Field	Required content
Test ID / revision	Stable identifier linked to requirement C1-C10
Build identity	Git commit, image/BSP version, configuration hash
Equipment	Model, serial number, firmware, calibration status where relevant
Topology	Ports, cables, addresses, baud/parity, VLAN/interface binding
Stimulus	Exact values and fault timing
Expected result	State, output, alarm, log, timing limit
Actual result	Captures, logs, contact measurement, screenshots
Verdict	PASS / FAIL / BLOCKED with defect reference

Prefer machine-readable test vectors (JSON/YAML/CSV) plus an automated runner for L1/L2. Preserve packet captures, serial traces, and controller logs as evidence artifacts.

10.1.12 To do

1. Author the PF2 state-machine specification: thresholds, hysteresis, debounce, stale-data timeout, priorities, and safe states.
2. Create a versioned Modbus meter register-map abstraction and `pymodbus` scenario runner.
3. Build L1 test vectors for normal ramps, boundary values, timeout, exceptions, malformed data, and restart.
4. Buy two known-quality isolated USB-to-RS-485 adapters for engineering use.
5. Obtain the selected real analyzer's official Modbus register map.
6. Define DI/DO safe states before energizing the carrier test box.
7. Add IEC 61850 SCL/test model and IEC 104 interoperability cases.
8. Define which L4 scenarios justify external HIL cost after L1-L3 results.

10.1.13 RAG keywords

validation ladder, PF2 bench, Modbus simulator, pymodbus, hardware in the loop, HIL, RS-485, energy analyzer, bad CRC, meter timeout, curtailment, DI test box, dry contact, IEC 61850, GOOSE, MMS, IEC 104, safe state, fault injection, ccli-validation-strategy

Chapter 11

Mocking / PF2 bench BOM

11.1 CCI Mocking / PF2 Bench BOM

Document ID: CCLI-BOM-BENCH-001 **Revision:** 1.0 **Date:** 2026-07-25 **RAG source_id:** ccli-mocking-bench-bom **Companion strategy:** CCI_Validation_and_Mocking_Strategy.md (ccli-validation) **Companion product BOM:** CCI_Prototype_BOM.md (field CCI parts - separate budget) **Purpose:** Buy list for the progressive PF2 validation bench (L1 software mock -> L2 RS-485/Ethernet HIL -> L3 real analyzer). L4 power-system HIL is listed as deferred / study-only. **Scope rule:** This BOM is for the **lab bench**, not the field CCI carrier. The Verdin kit/SoM appears here only as the **controller under test** (already on the product Phase 1 buy list). Status keys: **BUY NOW** · **BUY NEXT** · **DEFER** · **HAVE** · **TBD** (model not locked) · **OPTIONAL**

11.1.1 Buy waves (recommended)

Wave	Level	Goal	Est. incremental EUR	Decision
A	L1 + L2	Software mock + physical RS- 485/Ethernet HIL	180-350	BUY NOW (excl. PC / kit already owned)
B	L2+ C4	DI test box (24 V first) + DO lamps	80-180	BUY NEXT af- ter Wave A works
C	L3	Real energy analyzer + register map	400-1,200+	BUY NEXT before field pilot
D	L4	Typhoon / OPAL-RT / RTDS campaign	10k-100k+ or rental	DEFER until L1-L3 pass

Assumed already on product Phase 1 path (do not double-count):

Item	PN	Est. EUR	Notes
Verdin Development Board + HDMI	99991105	~300	Controller under test
Verdin iMX8M Plus Quad 4GB IT	0063	150-350	SoM under test
Engineering laptop/PC -		-	Host for simulators / Wireshark / clients

Wave A+B bench-only subtotal (typical): ~EUR 260-530 (adapters, switch, cabling, DI/DO

fixture - not including analyzer or kit).

11.1.2 Topology (what this BOM builds)

Engineering PC

```

+-- pymodbus / ModbusPal / QModMaster (L1)
+-- libiec61850 client / IEC104 test client
+-- Wireshark + serial capture
+-- USB-RS485 adapter(s) ---- RS-485 cable ---- Verdin kit RS485 DB9
+-- Ethernet ----- unmanaged switch --- Verdin kit Eth0/Eth1
                                     |
                                     +-- DI test box (24 V Wave B)
                                     +-- DO indicator lamps (Wave B)

```

Later (Wave C): replace/supplement Modbus emulator with a real analyzer on the same RS-485 bus.

11.1.3 Wave A - L1 software mock + L2 interface HIL (BUY NOW)

2.1 Software (zero or low HW cost)

Ref	Item	Qty	Est. EUR	Status	Notes
SW-MB	pymodbus simulator + scenario runner	1	0	BUY NOW	Preferred for scripted P/Q/PF2 ramps, timeout, exceptions
SW-MB2	ModbusPal / QModMaster / ModRSsim2	0-1	0	OPTIONAL	Manual UI debugging beside pymodbus
SW-61850	libiec61850 client tools	1	0	BUY NOW	GPLv3 for lab; commercial license before product ship
SW-104	IEC 60870-5-104 test client (open or licensed)	1	0-TBD	TBD	Pick after stack choice; lib60870 is one option
SW-CAP	Wireshark (+ USB-serial capture if needed)	1	0	BUY NOW	Eth_A/Eth_B and GOOSE/MMS evidence
SW-TV	Versioned test-vector pack (JSON/YAML)	1	eng time	BUY NOW	L1-P-01..04, L1-C-01..03, PF2 sequence from strategy doc

2.2 Serial / Modbus physical (L2)

Ref	Item	Suggested class / example	Qty	Est. EUR	Status	Doc
A-USB485	Isolated USB-to-RS-485 adapter	FTDI/CH340 industrial isolated (e.g. Waveshare / DSD TECH / Moxa UPort class)	2	25-60 ea	BUY NOW	Prefer isolated ; buy 2 so one can stay on bus while second injects faults
A-CAB485	RS-485 twisted-pair cable	Belden 3105A-class or CAT5e pair A/B + GND	10 m	15-30	BUY NOW	Mark A/B/GND; keep a spare 2 m jumper
A-TERM	120 ohm termination + bias kit	Plug-in 120R, optional 680R bias	2 sets	5-15	BUY NOW	For terminate / unterminate fault cases
A-DB9	DB9 male/female adapters / breakout	Matches Verdin Dev Board RS485 DB9	1-2	8-20	BUY NOW	Kit has 1x RS485 DB9 - not opto RJ45
A-POL	A/B polarity switch or screw-terminal block	DIN rail terminal + label	1	5-15	BUY NOW	Controlled A/B reverse tests

2.3 Ethernet / protocol (L2)

Ref	Item	Suggested class	Qty	Est. EUR	Status	Notes
A-SW	Unmanaged GbE switch (5-8 port)	TP-Link / Netgear / industrial DIN if available	1	20-60	BUY NOW	Separate Eth_A / Eth_B / plant / PC ports with labels
A-ETH	CAT6 patch leads (labelled)	Short 0.5-2 m	6	10-20	BUY NOW	Labels: Eth_A, Eth_B, Plant1, Plant2, PC, Spare
A-TAP	Optional Ethernet TAP / mirror	Soft (Wire-shark) first; hardware TAP later	0-1	0-150	OPTIONAL	Soft capture enough for Wave A

2.4 Lab power / misc (Wave A)

Ref	Item	Qty	Est. EUR	Status	Notes
A-PSU24	Bench PSU 24 Vdc (shared with DI box later)	1	40-90	BUY NOW if not owned	Start DI tests at 24 V only
A-ESD	ESD mat / wrist strap	1	15-30	OPTIONAL	Recommended for SoM handling
A-LAB	Label printer or cable tags	1	10-25	BUY NOW	Non-negotiable for fault-injection clarity

Wave A hardware subtotal (typical): ~EUR 180-350 (excluding existing PC and Toradex kit).

11.1.4 Wave B - DI test box + DO lamps (BUY NEXT)

Product target remains **DI 10-120 Vdc x5** and **DO relay x3**. On the **eval kit**, DI/DO are GPIO/stub only - Wave B either:

1. exercises **application mock DI/DO** via USB-GPIO / relay hat until carrier exists, or
2. waits for **custom carrier** and then uses the real field DI/DO circuits.

Recommendation: build a **24 V guarded fixture now** (safe), design the 48/110 V paths for the carrier, and do not energize 110 Vdc until carrier input design review passes.

3.1 DI fixture (24 V first)

Ref	Item	Qty	Est. EUR	Status	Notes
B-PSU24	24 Vdc PSU (if not A-PSU24)	1	-	HAVE if Wave A bought	Fuse both rails
B-PSU48	48 Vdc PSU	0-1	40-80	DEFER	After carrier DI design freeze
B-PSU110	110 Vdc source	0-1	TBD	DEFER	Hazardous - competent person + current limit + E-stop
B-SW	Selector / toggle switches (named states)	5	10-25	BUY NEXT	Breaker Closed/Open, Alarm, Meter Fault, Permissive
B-FUSE	Fuses + holders (per channel / set supply)		10-20	BUY NEXT	Current-limit every injection path
B-TB	DIN terminal blocks + DIN rail + enclosure	1 kit	25-50	BUY NEXT	Guarded terminals; no bare 110 V banana plugs
B-ESTOP	Emergency disconnect / key switch	1	15-30	BUY NEXT	Required before any >24 V
B-LED	Status LEDs on fixture	5	5-10	OPTIONAL	Show injected DI state

Named DI stimuli (map in software ICD later):

Channel label	Meaning
DI1	Breaker Closed
DI2	Breaker Open
DI3	Alarm Active
DI4	Meter Fault
DI5	External Permissive

3.2 DO observation

Ref	Item	Qty	Est. EUR	Status	Notes
B-LAMP	24 V indicator lamps / LED panels	3	10-25	BUY NEXT	One per PF2 / command / spare DO
B-PLCIN	Optional isolated DI module (PLC input)	0-1	40-120	OPTIONAL	Cleaner bounce/timing capture than lamps alone
B-SCOPE	Portable scope / logic analyzer	0-1	0-TBD	OPTIONAL	Contact bounce / response time

Wave B subtotal (24 V fixture): ~EUR 80-180.

11.1.5 Wave C - L3 real energy analyzer (BUY NEXT before field pilot)

Do **not** buy the analyzer until Wave A Modbus master + PF2 vectors pass on the simulator.

Ref	Item	Example families	Qty	Est. EUR	Status	Blocks
C-AN	Energy analyzer (Modbus RTU)	Schneider PM / Janitza UMG / Carlo Gavazzi / Siemens PAC	1	400-1,200+	TBD lock model	L3 gate
C-MAP	Official Modbus register map PDF	Vendor portal for chosen model	1	0	MISSING until model lock	Software + ingest ccli-modbus-analyzer-map
C-PS	Analyzer supply + VT/CT test path	Per vendor	1	TBD	TBD	May need secondary injection; not always "type registers by hand"
C-CAB	RS-485 plant cable to analyzer	Same class as A-CAB485	1	-	reuse	Match product pinout when carrier exists

L3 acceptance: map on disk + ingested; every consumed register has address/type/scale/unit; readings agree with analyzer display within tolerance; disconnect/reboot cases pass.

11.1.6 Wave D - L4 power-system HIL (DEFER)

Ref	Item	Qty	Est. EUR	Status	Notes
D-HIL	Typhoon HIL / OPAL-RT / RTDS access	campaign	rental or 10k-100k+	DEFER	Closed-loop grid/inverter/protection
D-IED	Optional IED / GOOSE publisher HW	TBD	TBD	DEFER	Only if L4 scope needs physical IEDs
D-OMICRON	Omicron IEDScout / Station-Scout	0-1	license	OPTIONAL	Strong for 61850; not required to start L1/L2

L4 is **not** on the Phase 1 EUR 1,000 product prototype budget.

11.1.7 Controller under test (already on product BOM)

Ref	Item	PN	Qty	Est. EUR	Status	Bench role
U-KIT	Verdin Development Board + HDMI	99991105	1	~300	product BOM	L2 DUT: 2x GbE, RS485 DB9
U-SOM	Verdin iMX8M Plus Quad 4GB IT	0063	1	150-350	product BOM	Compute / HAB / BSP
U-PSU	Kit 7-24 V supply (brick - or bench)		1	15-40	BUY NOW (if not in kit)	Separate from DI PSU

C2 reminder: kit RS485 is **not** product C2 (need 2x opto RJ45 on carrier). Wave A/B prove software + drivers; repeat on carrier.

11.1.8 Consolidated buy list (print / purchase)

7.1 Order this week (Wave A)

Priority	Ref	Item	Qty	Est. EUR
1	A-USB485	Isolated USB-RS485 adapters	2	50-120
2	A-SW	5-8 port GbE switch	1	20-60
3	A-CAB485 + A-ETH + A-DB9 + A-TERM	Cables / DB9 / termination	1 kit	40-80
4	A-PSU24	24 V bench PSU	1	40-90
5	A-LAB	Cable labels	1	10-25
Soft	SW-MB + SW-61850 + SW-CAP + SW-TV	Simulators + vectors	-	0 + eng

7.2 Order after first Modbus poll works (Wave B)

Priority	Ref	Item	Qty	Est. EUR
1	B-TB + B-SW + B-FUSE + B-ESTOP	24 V DI fixture kit	1	60-125
2	B-LAMP	DO indicator lamps x3	3	10-25

7.3 Order before field pilot (Wave C)

Priority	Ref	Item	Qty	Est. EUR
1	C-AN	Lock analyzer model + buy	1	400-1,200+
2	C-MAP	Download + ingest register map	1	0

11.1.9 Cost roll-up

Bucket	Est. EUR	In product EUR 1k budget?
Wave A bench HW	180-350	No - lab tooling
Wave B DI/DO fixture	80-180	No - lab tooling
Wave C analyzer	400-1,200+	No - lab/plant instrument
Toradex kit + SoM	450-650	Yes - product Phase 1
Wave D HIL	rental / capital	No - later campaign

Treat Waves A-C as a **lab CapEx** line separate from the ~EUR 1,000 single-unit CCI prototype BOM.

11.1.10 Documents still required for this bench

Doc	Status	source_id / path
Validation & mocking strategy	HAVE / INGESTED	ccli-validation-strategy
This bench BOM	HAVE	ccli-mocking-bench-bom
PF2 state-machine spec	MISSING	ccli-app-pf2 (planned)
pymodbus scenario pack	MISSING	repo under 06-application/
Analyzer Modbus map	MISSING	ccli-modbus-analyzer-map
USB-RS485 adapter DS (chosen MPN)	MISSING	08-engineering/reference/
DI fixture wiring diagram	MISSING	author with Wave B

11.1.11 Mapping to C1-C10 / validation levels

Control	Bench coverage
C1 Eth_A/B	Wave A switch + dual kit GbE (Partial vs product 4-port)
C2 RS-485	Wave A/B L2 on kit DB9; product 2x opto still on carrier
C3 IEC 61850	Wave A clients on Ethernet; L4 for closed-loop GOOSE
C4 DI/DO	Wave B 24 V fixture; full 10-120 Vdc on carrier
C5 GNSS	Fault-inject "GPS lost" in SW first; HW later
C6 secure boot	Not this BOM - SoM/HAB path on product BOM
C10 power	Kit PSU for DUT; separate fused PSU for DI injection

11.1.12 To do

1. Freeze **A-USB485** MPN (isolated, Linux `ttUSB` stable) and download DS.
2. Author PF2 state machine + `tests/modbus/l1_vectors.yaml`.
3. Build Wave A; run L1-P/C vectors end-to-end.
4. Build 24 V DI fixture; **do not** buy 110 V source yet.
5. Shortlist 2 analyzer models; obtain register maps before purchase.
6. Keep L4 as a separate CapEx decision after L1-L3 evidence pack exists.

11.1.13 Keywords

mocking bench bom, PF2 bench, USB-RS485, DI test box, DO lamp, energy analyzer, Wave A, Wave B, Wave C, pymodbus, 99991105, ccli-mocking-bench-bom

VI RAG knowledge-base

Chapter 12

RAG knowledge-base (clustered inventory)

12.1 CCI RAG Knowledge Base - Clustered Inventory

Document ID: CCLI-KB-RAG-001 **Revision:** 1.0 **Date:** 2026-07-25 **RAG source_id:** ccli-rag-knowledge-base
Purpose: Single clustered view of the document gate, KB branches, ingest status, and open items for Architecture Freeze and RAG completeness. **Companions:** CCI_Prototype_BOM.md (detail BOM) | DOWNLOADS.md | README.md (branch briefs)

12.1.1 Status rollup (document gate D0-D21)

Cluster	Items	HAVE	INGESTED	PARTIAL	MISSING	Gate
A Programme & architecture	3	3	0	0	0	Ready
B Architecture freeze blockers	2	0	0	0	2	Blocks schematic
C SoM & silicon	4	2	2	0	2	PCN + IEC DS open
D Carrier electronics (MPN TBD)	5	0	0	1	4	After pin budget
E Plant & protocols	3	2	2	0	1	Modbus map open
F Optional / rev A	4	0	0	1	3	Not Phase 1 block
G Fab & mechanical	2	0	0	0	2	After schematic
Total	23	7	4	2	14	

Architecture Freeze today: Cluster **B** (pin budget + carrier block diagram) and **C** (PCN letter + IMX8MPIEC) are the critical path. Cluster **A** is complete.

12.1.2 Cluster A - Programme & architecture (ready)

#	Document	Status	source_id / path	Why
D0	Architecture block diagram	HAVE	CCI_Architecture.png / mermaid	Functional interfaces
D1	Project roadmap + BOM cost envelope	HAVE	ccli-project-roadmap	Budget / phases
D2	Vendor selection + C1-C10 scorecard	HAVE	ccli-vendor-selection	Kilobits carrier split

12.1.3 Cluster B - Architecture freeze blockers (P0)

#	Document	Status	source_id / path	Why
D3	Pin budget / connector map	MISSING	CCI_Pin_Budget.md -> ccli-pin-budget	Every SoM ball -> net -> connector
D4	Carrier block diagram (frozen)	MISSING	CCI_Carrier_BlockDiagram.md -> ccli-carrier-block	C1/G2/G4/G5 freeze gate

12.1.4 Cluster C - SoM & silicon

#	Document	Status	source_id / path	Why
D5	Verdin Plus IT SoM datasheet (PN 0063)	HAVE / INGESTED	Toradex_Verdin_iMX8M_Plus_ITk_Datasheet.pdf -> ccli-toradex-verdin-plus-som	Verdin Plus ITk
D5a	NXP i.MX 8M Plus CEC SoC DS	HAVE / INGESTED	reference/IMX8MPCSoC.pdf -> ccli-nxp-imx8m-plus-cec	SoC pdf
D5b	NXP i.MX 8M Plus IEC SoC DS	MISSING	reference/IMX8MPCSoC.pdf -> ccli-nxp-imx8m-plus-iec	Industrial-temp silicon claims
D6	Longevity / PCN letter (SKU 0063)	MISSING	vendor PDF	10-year industrial path

Related on disk (not in D-table): Verdin dev board DS, carrier guide, family spec - all **INGESTED**.

12.1.5 Cluster D - Carrier electronics (MPN TBD)

#	Document	Status	source_id / path	Why
D11	Relay + opto datasheets (DI/DO)	MISSING	per MPN	C4 field 10-120 Vdc / 250 Vac DO
D12	Ethernet switch / PHY datasheets	MISSING	per MPN	2 plant ports beyond SoM GbE
D13	Isolated RS-485 transceiver DS	MISSING	per MPN	2x plant Modbus
D14	Power (12-24 Vdc -> SoM rails) DS	MISSING	buck/LDO TBD	C10 product path
D19	EMC / connector family (RJ45 8/8)	PARTIAL	CEI manual HAVE ; part PNs TBD	Plant serial pinout

12.1.6 Cluster E - Plant & protocols

#	Document	Status	source_id / path	Why
D8	Energy analyzer Modbus register map	MISSING	07-protocols/... -> ccli-modbus-analyzer-map	RS-485 plant serial
D20	Variscite DART + DT8M board (alternate)	HAVE / INGESTED	reference/vendor- Dauid cable/alternate	Dauid cable/alternate
D21	Compulab IOT-GATE ref (study)	HAVE / INGESTED	reference/vendor- by Q1 interface	by Q1 interface

12.1.7 Cluster F - Optional / rev A

#	Document	Status	Notes
D9	GNSS module datasheet	MISSING	C5 - pick MPN after carrier
D10	LTE module datasheet	PARTIAL	Optional rev A
D15	TPM / secure element DS	MISSING	If not on SoM SKU
D16	Anti-tamper switch + accelerometer	MISSING	Product / CRA
D17	OLED / HMI DS	MISSING	Optional proto

12.1.8 Cluster G - Fab & mechanical

#	Document	Status	source_id / path	Why
D7	ActiveBOM export (Altium)	MISSING	after schematic	Fab-ready MPN list
D18	Enclosure / DIN-rail mechanical	MISSING	drawing + BOM	Not eval board form

12.1.9 KB branch clusters (RAG tree)

Branch	Folder	RAG readiness	Key source_ids
Yocto Docs	01-yocto/	Strong	ccli-yocto-overview, ccli-nxp-yocto-ug, ccli-github-yocto-refs
BitBake Docs	02-bitbake/	Strong	ccli-bitbake-manual
Custom Layers	03-custom-layers/	Missing	ccli-meta-ccli-readme (planned)
Recipes	04-recipes/	Missing	ccli-recipes-index (planned)
Kernel & DT	05-kernel-dt/	Partial	SoC DS ingested; car- rier overlay missing
Application	06-application/	Missing	ccli-app-iec61850, ccli-app-modbus, ccli-app-pf2
Protocols	07-protocols/	Strong	CEI O/T, manual, regs class, EU CRA/RED
Engineering	08-engineering/	Strong	vendor, BOM, vali- dation, bench BOM, Toradex PDFs

12.1.10 GitHub / layer-pin clusters (SHA policy)

Cluster	Repos	Status
A Harvested	imx-manifest, meta-freescale, li- biec61850/60870, toradex signed- boot, meta-secure-core	INGESTED via ccli-github-yocto-refs
B Toradex lead	toradex-manifest, meta-toradex- security, build-signed-imx8m- bootloader	manifest + security PIN TBD
C NXP silicon	imx-manifest, meta-imx, meta- freescale, mfgtools	meta-imx LISTED
D-F Alternates	Variscite platform, Compulab meta-bsp-imx8mm	LISTED
G Security / IMA	meta-toradex-security, Wind-River meta-secure-core, meta-security	PIN TBD
H Protocols	libiec61850, lib60870, pymodbus (bench)	61850/60870 INGESTED

Canonical catalog: 01-yocto/CCI_GitHub_Paths_and_Layer_Pins.md (ccli-github-layer-pins)
Freeze artifact: CCI_Layer_Pinlist.json -> ccli-layer-pinlist (**MISSING**)

12.1.11 Ingest clusters (Telemetry)

Cluster	Count (approx.)	Examples
Italian grid (CEI)	3	cei-0-16-allegato-o, -t, manual
Programme docs	5+	roadmap, vendor selection, proto-type BOM, validation
Toradex / vendor PDFs	7+	plus-som, devboard, carrier guide, Variscite, Compulab study
NXP silicon	3	IMX8MPCEC, IMX8MDQLQCEC, precise extract
Yocto / BitBake	4+	mega-manual, UG10164, github refs, layer pins
Reference patterns	3	Moxa, iGate, STM32
EU regulations	2	RED, CRA

Rebuild ingest: Telemetry System/scripts/ingest-ccli.ps1

12.1.12 To do (by cluster priority)

P0 - blocks Architecture Freeze

1. Author **D3** CCI_Pin_Budget.md
2. Author **D4** CCI_Carrier_Block_Diagram.md
3. Download **D5b** IMX8MPIEC.pdf
4. Request **D6** Toradex PCN for PN **0063**

P1 - blocks carrier schematic / L3 bench

1. Obtain **D8** energy analyzer Modbus map
2. Pick MPNs + DS for **D11-D14** (C4, C1, C2, C10)
3. Author PF2 spec + Modbus test vectors (ccli-app-pf2)

P2 - product / optional

1. Licensed norm PDFs (ARERA, 61557-12, 62443, 62351)
2. **D7** ActiveBOM after Altium
3. **D18** enclosure drawing

12.1.13 Keywords

rag knowledge base, document gate, cluster, D0, pin budget, carrier block, IMX8MPIEC, PCN, 0063, ingest, Architecture Freeze, ccli-rag-knowledge-base

Chapter 13

GitHub paths & layer pin catalog

13.1 CCI GitHub Paths & Layer Pin Catalog

Document ID: CCLI-GITHUB-PINS-001 **Revision:** 1.0 **Date:** 2026-07-25 **RAG source_id:** ccli-github-layer-pins **Companion harvest:** 01-yocto/CCI_GitHub_Yocto_References.md (ccli-github-
Purpose: One map of every upstream Git URL, branch/tag, and SHA pin policy for reproducible Yocto builds, HAB/signed-boot, and protocol stacks on the Toradex lead path. **Pin rule:** At Architecture Freeze, record **exact commit SHA** (or vendor release tag + manifest XML) for every layer in ccli-layer-pinlist - never float on HEAD for product builds. Status: **HAVE** = README harvested in github-refs/ and in RAG · **LISTED** = URL documented here only · **PIN TBD** = lock SHA at freeze · **MISSING** = not yet documented/harvested

13.1.1 SHA / hash recording template (Architecture Freeze)

Field	Example (Toradex lead)
Layer / repo	meta-toradex-nxp
Remote URL	https://github.com/toradex/... or git.toradex.com/...
Branch or tag	scarthgap-7.x.y / kirkstone-6.x.y
Commit SHA	abc1234... (from repo manifest or git rev-parse HEAD)
Manifest file	e.g. default.xml from toradex-manifest
BSP release	e.g. TorizonCore 7.x / Toradex Yocto 6.6.53
Recorded by / date	Engineering sign-off row

Where to store pins: knowledge-base/03-custom-layers/CCI_Layer_Pinlist.json (planned, ccli-layer-pinlist) - not created yet. **Security hashes (HAB / FIT / IMA):** bootloader and image **signatures** are not Git SHAs - track in secure-boot runbook (ccli-secure-boot-runbook, planned). Git repos below supply the **signing pipeline**, not the hash values themselves.

13.1.2 Cluster A - Already harvested (github-refs/ + RAG)

Local file	Upstream URL	Branch / note	RAG	Status
github-refs/nxp-imx-manifest	https://github.com/scarthgap/imx-manifest	imx-linux-scarthgap	github-yocto-refs	HAVE / IN-GESTED
github-refs/nxp-imx-manifest	https://github.com/scarthgap/imx-manifest	imx-linux-wrynosi (LF6.18.20 tip)	github-yocto-refs	HAVE / IN-GESTED
github-refs/meta-freescale-scarthgap	https://github.com/Freescale/meta-freescale-scarthgap	scarthgap (Yocto 5.0 LTS)	ccli-github-yocto-refs	HAVE / IN-GESTED
github-refs/libiec61850	https://github.com/mz-automation/libiec61850	master (pin tag at freeze)	ccli-github-yocto-refs	HAVE / IN-GESTED
github-refs/lib60870	https://github.com/mz-automation/lib60870	master	ccli-github-yocto-refs	HAVE / IN-GESTED
github-refs/meta-secure-core	https://github.com/jiazhang0/meta-secure-core	master - deprecated	ccli-github-yocto-refs	HAVE / IN-GESTED
github-refs/toradex-signed-boot	https://github.com/toradex/build-signed-imx8m-bootloader	scarthgap-7.x	ccli-github-yocto-refs	HAVE / IN-GESTED

Combined ingest path: knowledge-base/01-yocto/CCI_GitHub_Yocto_References.md

13.1.3 Cluster B - Toradex lead path (PN 99991105 + SoM 0063)

Repo	URL	Branch / tag (verify at order)	Role	Status
toradex-manifest	https://git.toradex.com/toradex-manifest.git	Toradex BSP line (e.g. 6.6.x / 7.x)	Top-level repo manifest - primary pin surface	LISTED / PIN TBD
meta-toradex-security	https://github.com/toradex/meta-toradex-security	scarthgap-7.x (match signed-boot repo)	UAB, FIT signing, OP-TEE, fTPM docs	LISTED / PIN TBD
build-signed-imx8m-bootloader	https://github.com/toradex/build-signed-imx8m-bootloader	scarthgap-7.x	Containerized signed U-Boot for Verdin iMX8MP	HAVE (see Cluster A)
meta-toradex-bsp-common	https://github.com/toradex/meta-toradex-bsp-common	match manifest	Shared Toradex BSP recipes	LISTED / PIN TBD
meta-toradex-demos	https://github.com/toradex/meta-toradex-demos	match manifest	Demos (optional)	LISTED
TorizonCore / Easy Installer	https://developer.toradex.com	release images	Field flash reference, not a git layer	LISTED
mfgtools (uuu)	https://github.com/nxp-imx/mfgtools	tag matching BSP	USB recovery flash (referenced by Toradex signed-boot README)	LISTED / PIN TBD

Docs to harvest next: meta-toradex-security README-secure-boot-imx.md, README-optee.md (into github-refs/). **Machine name for Verdin iMX8M Plus:** verdin-imx8mp (from signed-boot README).

13.1.4 Cluster C - NXP silicon reference (all vendors)

Repo	URL	Branch / manifest	Role	Status
imx-manifest	https://github.com/nxp-imx/imx-manifest	imx-linux-scarthgap or imx-linux-wrynose-6.6.52-2.2.2.xml	NXP BSP manifest XMLs (e.g. imx-linux-wrynose-6.6.52-2.2.2.xml)	HAVE (Cluster A)
meta-imx	https://github.com/nxp-imx/meta-imx	branch matching manifest	NXP i.MX recipes, HAB/CAAM integration	LISTED / PIN TBD
meta-freescale	https://github.com/Freescale/meta-freescale	scarthgap	SoC support layer	HAVE (Cluster A)
meta-freescale-3rdparty	https://github.com/Freescale/meta-freescale-3rdparty	scarthgap	3rd-party boards	LISTED
mfgtools	https://github.com/nxp-imx/mfgtools	release tag	Manufacturing flash tool	LISTED

CCI pin note: Upstream docs pin **Yocto 5.0.13 Scarthgap**; NXP BSP tip is **LF6.18.20 / Wrynose** - reconcile with **Toradex manifest** at SoM freeze, not NXP tip alone.

13.1.5 Cluster D - Variscite alternate (DART-MX8M-PLUS)

Repo	URL	Branch	Role	Status
variscite-bsp-platform	https://github.com/varigit/variscite-bsp-platform	Scarthgap / 6.6.y	Top-level Variscite BSP checkout	LISTED / PIN TBD
meta-variscite-imx	https://github.com/varigit/meta-variscite-imx	Scarthgap 6.6.y	i.MX recipes for DART/VAR-SOM	LISTED / PIN TBD
meta-variscite-bsp	https://github.com/varigit/meta-variscite-bsp	match platform	Machine configs	LISTED

13.1.6 Cluster E - Compulab alternate / study

Repo	URL	Branch	Role	Status
meta-bsp-imx8mm	https://github.com/compulab-yokneam/meta-bsp-imx8mm	vendor default	CL-SOM / IOT-GATE BSP	LISTED / PIN TBD
meta-compulab	https://github.com/compulab-yokneam/meta-compulab	vendor default	Compulab common layer	LISTED

13.1.7 Cluster F - Yocto core & OE (reference - usually pinned via manifest)

Repo	URL	Branch	Role	Status
openembedded-core (poky)	https://git.yoctoproject.org/openembedded-core	scarthgap	OE-core	LISTED / PIN TBD
meta-yocto (poky)	https://git.yoctoproject.org/meta-yocto	scarthgap	Poky distro	LISTED
meta-openembedded	https://github.com/openembedded/meta-openembedded	scarthgap	meta-oe, meta-python, meta-networking	LISTED
bitbake	https://git.openembedded.org/bitbake	2.8 (Scarthgap era)	Build engine	LISTED

Doc on disk: Yocto 5.0.13 mega-manual PDF - [ccli-yocto-overview](#) (**INGESTED**).

13.1.8 Cluster G - Security, TPM, IMA, and image integrity

Repo	URL	Branch	Role	Status
meta-toradex-security	https://github.com/toradex/meta-toradex-security	scarthgap-7.x.y	Lead: HAB/FIT for Verdin iMX8MP	LISTED / PIN TBD
meta-secure-core (Wind-River)	https://github.com/Wind-River/meta-secure-core	active	Successor to jiazhang0 fork; UEFI/MOK/TPM/IMA	LISTED / PIN TBD
meta-secure-core (jiazhang0)	https://github.com/jiazhang0/meta-secure-core	deprecated	IMA measurement hashes, MODSIGN - see Cluster A	HAVE / deprecated
meta-security	https://git.yoctoproject.org/meta-security	scarthgap	Yocto hardening, TPM recipes	LISTED / PIN TBD
meta-security-framework	https://github.com/Wind-River/meta-secure-security-framework	-	WR security framework	LISTED

CCI C6 mapping: SoM HAB + Toradex signed-boot + optional TPM - Git paths above; product key hierarchy is **not** in public repos.

13.1.9 Cluster H - Application / protocol (BitBake sources)

Repo	URL	Branch	Role	License	Status
libiec61850	https://github.com/mz-automation/libiec61850	pin tag	MMS, GOOSE, SV (C3 Eth_A)	GPLv3 + commercial	HAVE
lib60870	https://github.com/mz-automation/lib60870	pin tag	IEC 60870-5-101/104	GPLv3 + commercial	HAVE
pymodbus	https://github.com/pymodbus-dev/pymodbus	pin tag	L1/L2 Modbus bench simulator (host PC)	BSD	LISTED
meta-ccli	(project repo - TBD)	-	Carrier DT, PF2, distro policy	proprietary	MISSING

13.1.10 Cluster I - Bench / lab tools (non-Yocto)

Repo	URL	Role	Status
pymodbus	https://github.com/pymodbus-dev/pymodbus	Modbus TCP/RTU test server (Wave A bench)	LISTED
pymodbus simulator examples	exam- same repo /examples/	PF2 test vectors	LISTED

13.1.11 Quick lookup - lead path minimum pin set

Order of operations when Toradex kit **99991105** + SoM **0063** arrives:

1. [git.toradex.com/toradex-manifest.git](https://github.com/toradex/toradex-manifest.git) - record **default.xml** revision
2. All layers referenced by that manifest - record **SHA per project**
3. github.com/toradex/meta-toradex-security @ matching branch - record SHA
4. github.com/toradex/build-signed-imx8m-bootloader @ scarthgap-7.x.y - record SHA

5. `github.com/nxp-imx/meta-imx` - only if pulled by manifest (verify)
 6. `github.com/Freescale/meta-freescale` @ `scarthgap` - record SHA
 7. Application pins: `libiec61850`, `lib60870` - record **release tag** for recipes
-

13.1.12 To do

1. Harvest **meta-toradex-security** docs into `github-refs/meta-toradex-security.md`
2. Harvest **meta-imx** README into `github-refs/meta-imx-scarthgap.md`
3. Create `CCI_Layer_Pinlist.json` after first successful bitbake
4. Add **toradex-manifest** default.xml snapshot to `01-yocto/manifests/` (read-only copy)
5. Add **pymodbus** pin to bench BOM software row
6. Re-ingest `ccli-github-layer-pins` + refresh combined `ccli-github-yocto-refs` if harvest expands

13.1.13 Keywords

`github`, `layer pin`, `SHA`, `commit`, `manifest`, `toradex-manifest`, `meta-toradex-security`, `meta-imx`, `imx-manifest`, `scarthgap`, `HAB`, `signed boot`, `IMA`, `libiec61850`, `ccli-github-layer-pins`, `ccli-layer-pinli`.